

Matrices

Chapter 3 · All Questions and Solutions



Part 1 · Examples

Q

1

Representing
data as a
matrix



Consider the following information regarding the number of men and women workers in three factories I, II and III

	Men workers	Women workers
I	30	25
II	25	31
III	27	26

Represent the above information in the form of a 3×2 matrix. What does the entry in the third row and second column represent?

Substitute

The information is represented in the form of a 3×2 matrix as follows:

$$A = \begin{bmatrix} 30 & 25 \\ 25 & 31 \\ 27 & 26 \end{bmatrix}$$

Conclusion

The entry in the third row and second column represents the number of women workers in factory III.

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2

Possible
orders for a
given
element
count



If a matrix has 8 elements, what are the possible orders it can have?

Conclusion

We know that a matrix of order $m \times n$ has $m \times n$ elements.

We need to find all possible orders of a matrix with 8 elements.

For this, we find all ordered pairs of natural numbers whose product is 8.

Thus, all possible ordered pairs are (1, 8), (8, 1), (4, 2), (2, 4)

Hence, possible orders are 1×8 , 8×1 , 4×2 , 2×4

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3

Constructing
a matrix
from a
formula



Construct a 3×2 matrix whose elements are given by $a_{ij} = \frac{1}{2} |i-3 j|$.

Given

In general a 3×2 matrix is given by $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$.

Now

Key formula

$$a_{ij} = \frac{1}{2} |i-3 j|, i = 1, 2, 3 \text{ and } j = 1, 2.$$

Substitute

Therefore

$$a_{11} = \frac{1}{2} |1-3 \times 1| = 1 \quad a_{12} = \frac{1}{2} |1-3 \times 2| = \frac{5}{2}$$

$$a_{21} = \frac{1}{2} |2-3 \times 1| = \frac{1}{2} \quad a_{22} = \frac{1}{2} |2-3 \times 2| = 2$$

$$a_{31} = \frac{1}{2} |3-3 \times 1| = 0 \quad a_{32} = \frac{1}{2} |3-3 \times 2| = \frac{3}{2}$$

Conclusion

Hence the required matrix is given by $A = \begin{bmatrix} 1 & \frac{5}{2} \\ \frac{1}{2} & 2 \\ 0 & \frac{3}{2} \end{bmatrix}$.

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4

Solving for unknowns by matrix equality



If $\begin{bmatrix} x+3 & z+4 & 2y-7 \\ -6 & a-1 & 0 \\ b-3 & -21 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 6 & 3y-2 \\ -6 & -3 & 2c+2 \\ 2b+4 & -21 & 0 \end{bmatrix}$ Find the values of a, b, c, x, y and z.

Substitute

The given matrices are equal.

So their corresponding elements must be equal.

Comparing the corresponding elements, we get

$$x+3 = 0, z+4 = 6, 2y-7 = 3y-2$$

$$a-1 = -3, 0 = 2c+2 \quad b-3 = 2b+4,$$

Conclusion

Simplifying, we get

$$a = -2, b = -7, c = -1, x = -3, y = -5, z = 2$$

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Solving for unknowns by matrix equality



Find the values of a, b, c, and d from the following equation:

$$\begin{bmatrix} 2a+b & a-2b \\ 5c-d & 4c+3d \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ 11 & 24 \end{bmatrix}$$

Substitute

Since the two matrices are equal, their corresponding elements must be equal.

Equating them, we get

$$2a+b = 4, 5c-d = 11$$

$$a-2b = -3, 4c+3d = 24$$

Conclusion

Solving these equations, we get

$$a = 1, b = 2, c = 3 \text{ and } d = 4$$

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6

Addition of matrices



Given $A = \begin{bmatrix} \sqrt{3} & 1 & -1 \\ 2 & 3 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & \sqrt{5} & 1 \\ -2 & 3 & \frac{1}{2} \end{bmatrix}$, find A+B Since A, B are of the same order 2×3 . Therefore, addition of A and B is defined and is given by

$$A+B = \begin{bmatrix} 2+\sqrt{3} & 1+\sqrt{5} & 1-1 \\ 2-2 & 3+3 & 0+\frac{1}{2} \end{bmatrix} = \begin{bmatrix} 2+\sqrt{3} & 1+\sqrt{5} & 0 \\ 0 & 6 & \frac{1}{2} \end{bmatrix}$$

- Note 1. We emphasise that if A and B are not of the same order, then A+B is not

defined. For example if $A = \begin{bmatrix} 2 & 3 \\ 1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 0 & 1 \end{bmatrix}$, then A+B is not defined.

2. We may observe that addition of matrices is an example of binary operation on the set of matrices of the same order.

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7

Scalar
multiplication
and
subtraction
of matrices

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If $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix}$, then find 2 A-B.

Conclusion

We have

$$\begin{aligned} 2A - B &= 2 \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} - \begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 4 & 6 \\ 4 & 6 & 2 \end{bmatrix} + \begin{bmatrix} -3 & 1 & -3 \\ 1 & 0 & -2 \end{bmatrix} \\ &= \begin{bmatrix} 2-3 & 4+1 & 6-3 \\ 4+1 & 6+0 & 2-2 \end{bmatrix} = \begin{bmatrix} -1 & 5 & 3 \\ 5 & 6 & 0 \end{bmatrix} \end{aligned}$$

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8

Solving a
matrix
equation for
X

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If $A = \begin{bmatrix} 8 & 0 \\ 4 & -2 \\ 3 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -2 \\ 4 & 2 \\ -5 & 1 \end{bmatrix}$, then find the matrix X, such that $2A + 3X = 5B$.

Given

We have $2A + 3X = 5B$

or

$$2A + 3X - 2A = 5B - 2A$$

or

$$2A - 2A + 3X = 5B - 2A \quad (\text{Matrix addition is commutative})$$

or

$$O + 3X = 5B - 2A \quad (-2A \text{ is the additive inverse of } 2A)$$

or

$$3X = 5B - 2A \quad (O \text{ is the additive identity})$$

or

Key formula

$$X = \frac{1}{3} (5B - 2A)$$

Substitute

$$\begin{aligned} X &= \frac{1}{3} \left(5 \begin{bmatrix} 2 & -2 \\ 4 & 2 \\ -5 & 1 \end{bmatrix} - 2 \begin{bmatrix} 8 & 0 \\ 4 & -2 \\ 3 & 6 \end{bmatrix} \right) = \frac{1}{3} \left(\begin{bmatrix} 10 & -10 \\ 20 & 10 \\ -25 & 5 \end{bmatrix} + \begin{bmatrix} -16 & 0 \\ -8 & 4 \\ -6 & -12 \end{bmatrix} \right) \\ &= \frac{1}{3} \begin{bmatrix} 10-16 & -10+0 \\ 20-8 & 10+4 \\ -25-6 & 5-12 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} -6 & -10 \\ 12 & 14 \\ -31 & -7 \end{bmatrix} = \begin{bmatrix} -2 & -\frac{10}{3} \\ 4 & \frac{14}{3} \\ -\frac{31}{3} & -\frac{7}{3} \end{bmatrix} \end{aligned}$$

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9

Solving
simultaneous
matrix
equations



Find X and Y, if $X+Y = \begin{bmatrix} 5 & 2 \\ 0 & 9 \end{bmatrix}$ and $X-Y = \begin{bmatrix} 3 & 6 \\ 0 & -1 \end{bmatrix}$.

Given

$$\text{We have } (X+Y)+(X-Y) = \begin{bmatrix} 5 & 2 \\ 0 & 9 \end{bmatrix} + \begin{bmatrix} 3 & 6 \\ 0 & -1 \end{bmatrix}.$$

or

$$(X+X)+(Y-Y) = \begin{bmatrix} 8 & 8 \\ 0 & 8 \end{bmatrix} \Rightarrow 2X = \begin{bmatrix} 8 & 8 \\ 0 & 8 \end{bmatrix}$$

or

$$X = \frac{1}{2} \begin{bmatrix} 8 & 8 \\ 0 & 8 \end{bmatrix} = \begin{bmatrix} 4 & 4 \\ 0 & 4 \end{bmatrix}$$

Conclusion

Also

$$(X+Y)-(X-Y) = \begin{bmatrix} 5 & 2 \\ 0 & 9 \end{bmatrix} - \begin{bmatrix} 3 & 6 \\ 0 & -1 \end{bmatrix}$$

or

$$(X-X)+(Y+Y) = \begin{bmatrix} 5-3 & 2-6 \\ 0 & 9+1 \end{bmatrix} \Rightarrow 2Y = \begin{bmatrix} 2 & -4 \\ 0 & 10 \end{bmatrix}$$

or

$$Y = \frac{1}{2} \begin{bmatrix} 2 & -4 \\ 0 & 10 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 0 & 5 \end{bmatrix}$$

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10

Solving for
unknowns in
a matrix
equation



Find the values of x and y from the following equation:

$$2 \begin{bmatrix} x & 5 \\ 7 & y-3 \end{bmatrix} + \begin{bmatrix} 3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$$

Substitute

We have

$$2 \begin{bmatrix} x & 5 \\ 7 & y-3 \end{bmatrix} + \begin{bmatrix} 3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix} \Rightarrow \begin{bmatrix} 2x & 10 \\ 14 & 2y-6 \end{bmatrix} + \begin{bmatrix} 3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$$

or

$$\begin{bmatrix} 2x+3 & 10-4 \\ 14+1 & 2y-6+2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix} \Rightarrow \begin{bmatrix} 2x+3 & 6 \\ 15 & 2y-4 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}$$

or

$$2x+3 = 7 \text{ and } 2y-4 = 14 \text{ (Why?)}$$

or

$$2x = 7-3 \text{ and } 2y = 18$$

or

$$x = \frac{4}{2} \text{ and } y = \frac{18}{2}$$

Conclusion

i.e.

$$x = 2 \text{ and } y = 9.$$

Q

11

Matrix
addition and
subtraction -
rice sales

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Two farmers Ramkishan and Gurcharan Singh cultivates only three varieties of rice namely Basmati, Permal and Naura. The sale (in Rupees) of these varieties of rice by both the farmers in the month of September and October are given by the following matrices A and B. September Sales (in Rupees)

$$A = \begin{bmatrix} \text{Basmati} & \text{Permal} & \text{Naura} \\ 10,000 & 20,000 & 30,000 \\ 50,000 & 30,000 & 10,000 \end{bmatrix} \begin{matrix} \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix}$$

October Sales (in Rupees)

$$B = \begin{bmatrix} \text{Basmati} & \text{Permal} & \text{Naura} \\ 5000 & 10,000 & 6000 \\ 20,000 & 10,000 & 10,000 \end{bmatrix} \begin{matrix} \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix}$$

- (i) Find the combined sales in September and October for each farmer in each variety. (ii) Find the decrease in sales from September to October. (iii) If both farmers receive 2% profit on gross sales, compute the profit for each farmer and for each variety sold in October.

Given

- (i) The combined sales in September and October, for each farmer and each variety, are given by

Key formula

$$A+B = \begin{bmatrix} \text{Basmati} & \text{Permal} & \text{Naura} \\ 15,000 & 30,000 & 36,000 \\ 70,000 & 40,000 & 20,000 \end{bmatrix} \begin{matrix} \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix}$$

Substitute

- (ii) The change in sales from September to October is given by

Key formula

$$A-B = \begin{bmatrix} \text{Basmati} & \text{Permal} & \text{Naura} \\ 5000 & 10,000 & 24,000 \\ 30,000 & 20,000 & 0 \end{bmatrix} \begin{matrix} \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix}$$

(iii)

$$2\% \text{ of } B = \frac{2}{100} \times B = 0.02 \times B$$

$$= 0.02 \begin{bmatrix} \text{Basmati} & \text{Permal} & \text{Naura} \\ 5000 & 10,000 & 6000 \\ 20,000 & 10,000 & 10,000 \end{bmatrix} \begin{matrix} \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix}$$

$$= \begin{bmatrix} \text{Basmati} & \text{Permal} & \text{Naura} \\ 100 & 200 & 120 \\ 400 & 200 & 200 \end{bmatrix} \begin{matrix} \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix}$$

Conclusion

So, in October, Ramkishan gets ₹100, ₹200 and ₹120 as profit on each variety of rice, respectively.

Gurcharan Singh gets ₹400, ₹200 and ₹200 as profit on each variety of rice, respectively.

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12

Associativity
of matrix
multiplication



Find AB , if $A = \begin{bmatrix} 6 & 9 \\ 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 6 & 0 \\ 7 & 9 & 8 \end{bmatrix}$.

Conclusion

The matrix A has 2 columns.

This is equal to the number of rows of B .

So AB is defined.

Now

$$\begin{aligned} AB &= \begin{bmatrix} 6(2)+9(7) & 6(6)+9(9) & 6(0)+9(8) \\ 2(2)+3(7) & 2(6)+3(9) & 2(0)+3(8) \end{bmatrix} \\ &= \begin{bmatrix} 12+63 & 36+81 & 0+72 \\ 4+21 & 12+27 & 0+24 \end{bmatrix} = \begin{bmatrix} 75 & 117 & 72 \\ 25 & 39 & 24 \end{bmatrix} \end{aligned}$$

AB defined does not mean BA defined

Remark: If AB is defined, then BA need not be defined.

In the example above, AB is defined.

But BA is not defined, because B has 3 columns, while A has only 2 (and not 3) rows.

Suppose A and B are, respectively, $m \times n$, $k \times l$ matrices.

Then both AB and BA are defined if and only if $n = k$ and $l = m$.

In particular, if both A and B are square matrices of the same order, then both AB and BA are defined.

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Non-
commutativity
of matrix
multiplication



If $A = \begin{bmatrix} 1 & -2 & 3 \\ -4 & 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{bmatrix}$, then find AB , BA . Show that $A B \neq B A$.

Given

A is a 2×3 matrix, and B is a 3×2 matrix.

So AB and BA are both defined, and they are matrices of order 2×2 and 3×3 , respectively.

Note that

$$AB = \begin{bmatrix} 1 & -2 & 3 \\ -4 & 2 & 5 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 2-8+6 & 3-10+3 \\ -8+8+10 & -12+10+5 \end{bmatrix} = \begin{bmatrix} 0 & -4 \\ 10 & 3 \end{bmatrix}$$

and

$$BA = \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & 3 \\ -4 & 2 & 5 \end{bmatrix} = \begin{bmatrix} 2-12 & -4+6 & 6+15 \\ 4-20 & -8+10 & 12+25 \\ 2-4 & -4+2 & 6+5 \end{bmatrix} = \begin{bmatrix} -10 & 2 & 21 \\ -16 & 2 & 37 \\ -2 & -2 & 11 \end{bmatrix}$$

Conclusion

Clearly $AB \neq BA$

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Non-commutativity of matrix multiplication



If $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, then $AB = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. and

$$BA = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}. \text{ Clearly } AB \neq BA.$$

Thus matrix multiplication is not commutative.

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15

Zero product of non-zero matrices

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Find AB , if $A = \begin{bmatrix} 0 & -1 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 5 \\ 0 & 0 \end{bmatrix}$.

Conclusion

$$\text{We have } AB = \begin{bmatrix} 0 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 3 & 5 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

This shows that even if the product of two matrices is a zero matrix, it is not necessary that either matrix itself is a zero matrix.

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16

Verifying associativity of multiplication



If $A = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 0 & 3 \\ 3 & -1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 3 \\ 0 & 2 \\ -1 & 4 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 2 & 3 & -4 \\ 2 & 0 & -2 & 1 \end{bmatrix}$, find $A(BC)$, $(AB)C$ and show that $(AB)C = A(BC)$.

Given

$$\text{We have } AB = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 0 & 3 \\ 3 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 2 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} 1+0-1 & 3+2-4 \\ 2+0-3 & 6+0+12 \\ 3+0-2 & 9-2+8 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -1 & 18 \\ 1 & 15 \end{bmatrix}$$

$$(AB)C = \begin{bmatrix} 2 & 1 \\ -1 & 18 \\ 1 & 15 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & -4 \\ 2 & 0 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 2+2 & 4+0 & 6-2 & -8+1 \\ -1+36 & -2+0 & -3-36 & 4+18 \\ 1+30 & 2+0 & 3-30 & -4+15 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 4 & 4 & -7 \\ 35 & -2 & -39 & 22 \\ 31 & 2 & -27 & 11 \end{bmatrix}$$

Substitute

Now

$$BC = \begin{bmatrix} 1 & 3 \\ 0 & 2 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & -4 \\ 2 & 0 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 1+6 & 2+0 & 3-6 & -4+3 \\ 0+4 & 0+0 & 0-4 & 0+2 \\ -1+8 & -2+0 & -3-8 & 4+4 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & 2 & -3 & -1 \\ 4 & 0 & -4 & 2 \\ 7 & -2 & -11 & 8 \end{bmatrix}$$

Conclusion

Therefore

$$A(BC) = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 0 & 3 \\ 3 & -1 & 2 \end{bmatrix} \begin{bmatrix} 7 & 2 & -3 & -1 \\ 4 & 0 & -4 & 2 \\ 7 & -2 & -11 & 8 \end{bmatrix}$$

$$= \begin{bmatrix} 7+4-7 & 2+0+2 & -3-4+11 & -1+2-8 \\ 14+0+21 & 4+0-6 & -6+0-33 & -2+0+24 \\ 21-4+14 & 6+0-4 & -9+4-22 & -3-2+16 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 4 & 4 & -7 \\ 35 & -2 & -39 & 22 \\ 31 & 2 & -27 & 11 \end{bmatrix}. \text{ Clearly, } (AB) C = A(BC)$$

Q

17

Distributivity
of matrix
multiplication



If $A = \begin{bmatrix} 0 & 6 & 7 \\ -6 & 0 & 8 \\ 7 & -8 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix}$, $C = \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix}$ Calculate AC, BC and $(A+B) C$. Also, verify that $(A+B) C = AC+BC$

Given

$$\text{Now, } A+B = \begin{bmatrix} 0 & 7 & 8 \\ -5 & 0 & 10 \\ 8 & -6 & 0 \end{bmatrix}$$

So

$$(A+B) C = \begin{bmatrix} 0 & 7 & 8 \\ -5 & 0 & 10 \\ 8 & -6 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix} = \begin{bmatrix} 0-14+24 \\ -10+0+30 \\ 16+12+0 \end{bmatrix} = \begin{bmatrix} 10 \\ 20 \\ 28 \end{bmatrix}$$

Substitute

Further

$$AC = \begin{bmatrix} 0 & 6 & 7 \\ -6 & 0 & 8 \\ 7 & -8 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix} = \begin{bmatrix} 0-12+21 \\ -12+0+24 \\ 14+16+0 \end{bmatrix} = \begin{bmatrix} 9 \\ 12 \\ 30 \end{bmatrix}$$

and

$$BC = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix} = \begin{bmatrix} 0-2+3 \\ 2+0+6 \\ 2-4+0 \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \\ -2 \end{bmatrix}$$

So

$$AC+BC = \begin{bmatrix} 9 \\ 12 \\ 30 \end{bmatrix} + \begin{bmatrix} 1 \\ 8 \\ -2 \end{bmatrix} = \begin{bmatrix} 10 \\ 20 \\ 28 \end{bmatrix}$$

Conclusion

Clearly,

$$(A+B) C = AC+BC$$

Q

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Matrix
polynomial
identity



If $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & -2 & 1 \\ 4 & 2 & 1 \end{bmatrix}$, then show that $A^3 - 23A - 40I = O$

Given

$$\text{We have } A^2 = A \cdot A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & -2 & 1 \\ 4 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 3 & -2 & 1 \\ 4 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 19 & 4 & 8 \\ 1 & 12 & 8 \\ 14 & 6 & 15 \end{bmatrix}$$

Substitute

So

$$A^3 = A A^2 = \begin{bmatrix} 1 & 2 & 3 \\ 3 & -2 & 1 \\ 4 & 2 & 1 \end{bmatrix} \begin{bmatrix} 19 & 4 & 8 \\ 1 & 12 & 8 \\ 14 & 6 & 15 \end{bmatrix} = \begin{bmatrix} 63 & 46 & 69 \\ 69 & -6 & 23 \\ 92 & 46 & 63 \end{bmatrix}$$

Conclusion

Now

$$\begin{aligned} A^3 - 23A - 40I &= \begin{bmatrix} 63 & 46 & 69 \\ 69 & -6 & 23 \\ 92 & 46 & 63 \end{bmatrix} - 23 \begin{bmatrix} 1 & 2 & 3 \\ 3 & -2 & 1 \\ 4 & 2 & 1 \end{bmatrix} - 40 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 63 & 46 & 69 \\ 69 & -6 & 23 \\ 92 & 46 & 63 \end{bmatrix} + \begin{bmatrix} -23 & -46 & -69 \\ -69 & 46 & -23 \\ -92 & -46 & -23 \end{bmatrix} + \begin{bmatrix} -40 & 0 & 0 \\ 0 & -40 & 0 \\ 0 & 0 & -40 \end{bmatrix} \\ &= \begin{bmatrix} 63-23-40 & 46-46+0 & 69-69+0 \\ 69-69+0 & -6+46-40 & 23-23+0 \\ 92-92+0 & 46-46+0 & 63-23-40 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O \end{aligned}$$

Q

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Matrix
multiplication
- campaign
costs

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In a legislative assembly election, a political group hired a public relations firm to promote its candidate in three ways: telephone, house calls, and letters. The cost per contact (in paise) is given in matrix A as

$$A = \begin{bmatrix} \text{Cost per contact} \\ 40 \\ 100 \\ 50 \end{bmatrix} \begin{matrix} \text{Telephone} \\ \text{Housecall} \\ \text{Letter} \end{matrix}$$

The number of contacts of each type made in two cities X and Y is given by Telephone Housecall Letter $B = \begin{bmatrix} 1000 & 500 & 5000 \\ 3000 & 1000 & 10,000 \end{bmatrix} \begin{matrix} \rightarrow X \\ \rightarrow Y \end{matrix}$.

Find the total amount spent by the group in the two cities X and Y.

Substitute

We have

$$\begin{aligned} BA &= \begin{bmatrix} 40,000+50,000+250,000 \\ 120,000+100,000+500,000 \end{bmatrix} \begin{matrix} \rightarrow X \\ \rightarrow Y \end{matrix} \\ &= \begin{bmatrix} 340,000 \\ 720,000 \end{bmatrix} \begin{matrix} \rightarrow X \\ \rightarrow Y \end{matrix} \end{aligned}$$

Conclusion

So the total amount spent by the group in the two cities is 340,000 paise and 720,000 paise, i.e., ₹3400 and ₹ 7200, respectively.

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Properties of
transpose

(())

If $A = \begin{bmatrix} 3 & \sqrt{3} & 2 \\ 4 & 2 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 & 2 \\ 1 & 2 & 4 \end{bmatrix}$, verify that (i) $(A')' = A$, (ii) $(A+B)' = A' + B'$, (iii) $(k B)' = k B'$, where k is any constant.

Given

(i) We have

$$A = \begin{bmatrix} 3 & \sqrt{3} & 2 \\ 4 & 2 & 0 \end{bmatrix} \Rightarrow A' = \begin{bmatrix} 3 & 4 \\ \sqrt{3} & 2 \\ 2 & 0 \end{bmatrix} \Rightarrow (A')' = \begin{bmatrix} 3 & \sqrt{3} & 2 \\ 4 & 2 & 0 \end{bmatrix} = A$$

Thus $(A')' = A$

Substitute

(ii) We have

$$A = \begin{bmatrix} 3 & \sqrt{3} & 2 \\ 4 & 2 & 0 \end{bmatrix}, B = \begin{bmatrix} 2 & -1 & 2 \\ 1 & 2 & 4 \end{bmatrix} \Rightarrow A+B = \begin{bmatrix} 5 & \sqrt{3}-1 & 4 \\ 5 & 4 & 4 \end{bmatrix}$$

Therefore

$$(A+B)' = \begin{bmatrix} 5 & 5 \\ \sqrt{3}-1 & 4 \\ 4 & 4 \end{bmatrix}$$

Now

$$A' = \begin{bmatrix} 3 & 4 \\ \sqrt{3} & 2 \\ 2 & 0 \end{bmatrix}, B' = \begin{bmatrix} 2 & 1 \\ -1 & 2 \\ 2 & 4 \end{bmatrix},$$

So

$$A' + B' = \begin{bmatrix} 5 & 5 \\ \sqrt{3}-1 & 4 \\ 4 & 4 \end{bmatrix}$$

Thus

$$(A+B)' = A' + B'$$

Conclusion

(iii) We have

$$k B = k \begin{bmatrix} 2 & -1 & 2 \\ 1 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 2k & -k & 2k \\ k & 2k & 4k \end{bmatrix}$$

Then

$$(k B)' = \begin{bmatrix} 2k & k \\ -k & 2k \\ 2k & 4k \end{bmatrix} = k \begin{bmatrix} 2 & 1 \\ -1 & 2 \\ 2 & 4 \end{bmatrix} = k B'$$

Thus

$$(k B)' = k B'$$

Q

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Transpose of
a matrix
product



If $A = \begin{bmatrix} -2 \\ 4 \\ 5 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 3 & -6 \end{bmatrix}$, verify that $(AB)' = B' A'$.

Given



We have

$$A = \begin{bmatrix} -2 \\ 4 \\ 5 \end{bmatrix}, B = \begin{bmatrix} 1 & 3 & -6 \end{bmatrix}$$

then

$$AB = \begin{bmatrix} -2 \\ 4 \\ 5 \end{bmatrix} \begin{bmatrix} 1 & 3 & -6 \end{bmatrix} = \begin{bmatrix} -2 & -6 & 12 \\ 4 & 12 & -24 \\ 5 & 15 & -30 \end{bmatrix}$$

Now

$$A' = \begin{bmatrix} -2 & 4 & 5 \end{bmatrix}, B' = \begin{bmatrix} 1 \\ 3 \\ -6 \end{bmatrix}$$

$$B' A' = \begin{bmatrix} 1 \\ 3 \\ -6 \end{bmatrix} \begin{bmatrix} -2 & 4 & 5 \end{bmatrix} = \begin{bmatrix} -2 & 4 & 5 \\ -6 & 12 & 15 \\ 12 & -24 & -30 \end{bmatrix} = (AB)'$$

Conclusion

Clearly

Key formula

$$(AB)' = B' A'$$



Q

22

Matrix as
sum of
symmetric
and skew-
symmetric
parts



Express the matrix $B = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ as the sum of a symmetric and a skew symmetric matrix.

Given

Here

$$B' = \begin{bmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ -4 & 4 & -3 \end{bmatrix}$$

Substitute

Let

$$P = \frac{1}{2} (B + B') = \frac{1}{2} \begin{bmatrix} 4 & -3 & -3 \\ -3 & 6 & 2 \\ -3 & 2 & -6 \end{bmatrix} = \begin{bmatrix} 2 & \frac{-3}{2} & \frac{-3}{2} \\ \frac{-3}{2} & 3 & 1 \\ \frac{-3}{2} & 1 & -3 \end{bmatrix},$$

Now

$$P' = \begin{bmatrix} 2 & \frac{-3}{2} & \frac{-3}{2} \\ \frac{-3}{2} & 3 & 1 \\ \frac{-3}{2} & 1 & -3 \end{bmatrix} = P$$

Thus

$$P = \frac{1}{2} (B + B') \text{ is a symmetric matrix.}$$

Also, let

$$Q = \frac{1}{2} (B - B') = \frac{1}{2} \begin{bmatrix} 0 & -1 & -5 \\ 1 & 0 & 6 \\ 5 & -6 & 0 \end{bmatrix} = \begin{bmatrix} 0 & \frac{-1}{2} & \frac{-5}{2} \\ \frac{1}{2} & 0 & 3 \\ \frac{5}{2} & -3 & 0 \end{bmatrix}$$

Then

$$Q' = \begin{bmatrix} 0 & \frac{1}{2} & \frac{5}{3} \\ \frac{-1}{2} & 0 & -3 \\ \frac{-5}{2} & 3 & 0 \end{bmatrix} = -Q$$

Thus

$$Q = \frac{1}{2} (B - B') \text{ is a skew symmetric matrix.}$$

Now

$$P + Q = \begin{bmatrix} 2 & \frac{-3}{2} & \frac{-3}{2} \\ \frac{-3}{2} & 3 & 1 \\ \frac{-3}{2} & 1 & -3 \end{bmatrix} + \begin{bmatrix} 0 & \frac{-1}{2} & \frac{-5}{2} \\ \frac{1}{2} & 0 & 3 \\ \frac{5}{2} & -3 & 0 \end{bmatrix} = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} = B$$

Conclusion

Thus, B is written as the sum of a symmetric matrix and a skew symmetric matrix.

Q

23

**Matrix
power by
induction**

(())

If $A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$, then prove that $A^n = \begin{bmatrix} \cos n\theta & \sin n\theta \\ -\sin n\theta & \cos n\theta \end{bmatrix}$, $n \in \mathbb{N}$.

Given

We will prove this result using the principle of mathematical induction.

Substitute

First, we check the base case.

$$P(n): \text{ If } A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}, \text{ then } A^n = \begin{bmatrix} \cos n\theta & \sin n\theta \\ -\sin n\theta & \cos n\theta \end{bmatrix}, n \in \mathbb{N}$$

$$P(1): A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}, \text{ so } A^1 = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

So the result is true for $n = 1$.

Next, assume the result is true for $n = k$.

So

$$P(k): A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}, \text{ then } A^k = \begin{bmatrix} \cos k\theta & \sin k\theta \\ -\sin k\theta & \cos k\theta \end{bmatrix}$$

Conclusion

Now we prove that the result holds for $n = k+1$.

We get

$$\begin{aligned} A^{k+1} &= A \cdot A^k = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos k\theta & \sin k\theta \\ -\sin k\theta & \cos k\theta \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta \cos k\theta - \sin \theta \sin k\theta & \cos \theta \sin k\theta + \sin \theta \cos k\theta \\ -\sin \theta \cos k\theta + \cos \theta \sin k\theta & -\sin \theta \sin k\theta + \cos \theta \cos k\theta \end{bmatrix} \\ &= \begin{bmatrix} \cos (\theta+k\theta) & \sin (\theta+k\theta) \\ -\sin (\theta+k\theta) & \cos (\theta+k\theta) \end{bmatrix} = \begin{bmatrix} \cos (k+1)\theta & \sin (k+1)\theta \\ -\sin (k+1)\theta & \cos (k+1)\theta \end{bmatrix} \end{aligned}$$

So the result is true for $n = k+1$.

Thus, by the principle of mathematical induction, $A^n = \begin{bmatrix} \cos n\theta & \sin n\theta \\ -\sin n\theta & \cos n\theta \end{bmatrix}$ holds for all natural numbers.

Q

24

Symmetric product commuting matrices



If A and B are symmetric matrices of the same order, then show that AB is symmetric if and only if A and B commute, that is $AB = BA$.

Given

A and B are both symmetric matrices.

So $A' = A$ and $B' = B$.

Substitute

First, suppose AB is symmetric.

Then $(AB)' = AB$

But

$(AB)' = B' A' = BA$ (Why?)

So

$BA = AB$

Conclusion

Conversely, suppose $AB = BA$.

We now show that AB is symmetric.

Now

$(AB)' = B' A'$
 $= BA$ (as A and B are symmetric)
 $= AB$

Hence AB is symmetric.

Q

25

Solving a matrix equation for D



Let $A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 2 \\ 7 & 4 \end{bmatrix}$, $C = \begin{bmatrix} 2 & 5 \\ 3 & 8 \end{bmatrix}$. Find a matrix D such that $CD - AB = O$.

Given

A, B, C are all square matrices of order 2. Since $CD - AB$ is well defined, D must also be a square matrix of order 2.

Key formula

Let

$D = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then $CD - AB = O$ gives
 $\begin{bmatrix} 2 & 5 \\ 3 & 8 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} - \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 2 \\ 7 & 4 \end{bmatrix} = O$

Substitute

So

$\begin{bmatrix} 2a+5c & 2b+5d \\ 3a+8c & 3b+8d \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 43 & 22 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

or

$$\begin{bmatrix} 2a+5c-3 & 2b+5d \\ 3a+8c-43 & 3b+8d-22 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Comparing corresponding elements, we get

$$2a+5c-3 = 0$$

$$3a+8c-43 = 0$$

$$2b+5d = 0$$

$$3b+8d-22 = 0$$

Solving (1) and (2) gives $a = -191$, $c = 77$.

Solving (3) and (4) gives $b = -110$, $d = 44$.

Conclusion

So

$$D = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} -191 & -110 \\ 77 & 44 \end{bmatrix}$$

Q

1

Order and
elements of a
matrix



In the matrix $A = \begin{bmatrix} 2 & 5 & 19 & -7 \\ 35 & -2 & \frac{5}{2} & 12 \\ \sqrt{3} & 1 & -5 & 17 \end{bmatrix}$, write:

Part (i)

The order of the matrix,

Part (ii)

The number of elements,

Part (iii)

Write the elements a_{13} , a_{21} , a_{33} , a_{24} , a_{23} .

Conclusion

(i) Since, in the given matrix, the number of rows is 3 and the number of columns is 4, the order of the matrix is 3×4 .

(ii) Since the order of the matrix is 3×4 , there are $3 \times 4 = 12$ elements.

(iii) Here,

$$a_{13} = 19$$

$$a_{21} = 35$$

$$a_{33} = -5$$

$$a_{24} = 12$$

$$a_{23} = \frac{5}{2}$$

Q

2

Possible
orders for 24
or 13
elements

(())

If a matrix has 24 elements, what are the possible orders it can have? What, if it has 13 elements?

Given

We know that if a matrix is of the order $m \times n$, it has $m \times n$ elements.

Thus, to find all the possible orders of a matrix having 24 elements, we have to find all the ordered pairs of natural numbers whose product is 24.

The ordered pairs are: (1,24), (24,1), (2,12), (12,2), (3,8), (8,3), (4,6) and (6,4).

Conclusion

Hence, the possible orders of a matrix having 24 elements are:

$(1 \times 24), (24 \times 1), (2 \times 12), (12 \times 2), (3 \times 8), (8 \times 3), (4 \times 6)$ and (6×4) .

Substitute

(1,13) and (13,1) are the ordered pairs of natural numbers whose product is 13.

Conclusion

Hence, the possible orders of a matrix having 13 elements are (1×13) and (13×1) .

Q

3

Possible
orders for 18
or 5 elements



If a matrix has 18 elements, what are the possible orders it can have? What, if it has 5 elements?

Given

We know that if a matrix is of the order $m \times n$, it has $m \times n$ elements.

Thus, to find all the possible orders of a matrix having 18 elements, we have to find all the ordered pairs of natural numbers whose product is 18.

The ordered pairs are: (1,18), (18,1), (2,9), (9,2), (3,6) and (6,3).

Conclusion

Hence, the possible orders of a matrix having 18 elements are:

$(1 \times 18), (18 \times 1), (2 \times 9), (9 \times 2), (3 \times 6)$ and (6×3) .

Substitute

(1×5) and (5×1) are the ordered pairs of natural numbers whose product is 5.

Conclusion

Hence, the possible orders of a matrix having 5 elements are (1×5) and (5×1) .

Q**4**

Constructing
a 2×2 matrix
from a
formula



Construct a 2×2 matrix, $A = [a_{ij}]$, whose elements are given by:

Part (i)

$$a_{ij} = \frac{(i+j)^2}{2}$$

Part (ii)

$$a_{ij} = \frac{i}{j}$$

Part (iii)

$$a_{ij} = \frac{(i+2j)^2}{2}$$

Given

In general, a 2×2 matrix is given by $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$

$$(i) a_{ij} = \frac{(i+j)^2}{2}; i, j = 1, 2$$

Therefore,

$$a_{11} = \frac{(1+1)^2}{2} = \frac{4}{2} = 2$$

$$a_{12} = \frac{(1+2)^2}{2} = \frac{9}{2}$$

$$a_{21} = \frac{(2+1)^2}{2} = \frac{9}{2}$$

$$a_{22} = \frac{(2+2)^2}{2} = \frac{16}{2} = 8$$

Conclusion

Thus, the required matrix is

$$A = \begin{pmatrix} 2 & \frac{9}{2} \\ \frac{9}{2} & 8 \end{pmatrix}$$

Substitute

(ii)

$$a_{ij} = \frac{i}{j}; i, j = 1, 2$$

Therefore,

$$a_{11} = \frac{1}{1} = 1$$

$$a_{12} = \frac{1}{2}$$

$$a_{21} = \frac{2}{1} = 2$$

$$a_{22} = \frac{2}{2} = 1$$

Conclusion

Thus, the required matrix is

$$A = \begin{pmatrix} 1 & \frac{1}{2} \\ 2 & 1 \end{pmatrix}$$

Substitute

(iii)

$$a_{ij} = \frac{(i+2j)^2}{2} ; i, j = 1, 2$$

Therefore,

$$a_{11} = \frac{(1+2)^2}{2} = \frac{9}{2}$$

$$a_{12} = \frac{(1+4)^2}{2} = \frac{25}{2}$$

$$a_{21} = \frac{(2+2)^2}{2} = 8$$

$$a_{22} = \frac{(2+4)^2}{2} = 18$$

Conclusion

Thus, the required matrix is

$$A = \begin{pmatrix} \frac{9}{2} & \frac{25}{2} \\ 8 & 18 \end{pmatrix}$$

Q**5**

Constructing
a 3x4 matrix
from a
formula



Construct a 3×4 matrix, whose elements are given by:

Part (i)

$$a_{ij} = \frac{1}{2} |-3i+j|$$

Part (ii)

$$a_{ij} = 2i-j$$

Given



In general, a 3×4 matrix is given by

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{pmatrix}$$

(i) Given $a_{ij} = \frac{1}{2} |-3i+j| ; i = 1, 2, 3 \quad j = 1, 2, 3, 4$

$$a_{11} = \frac{1}{2} |-3(1)+1| = \frac{1}{2} |-3+1| = \frac{1}{2} |-2| = \frac{2}{2} = 1$$

$$a_{21} = \frac{1}{2} |-3(2)+1| = \frac{1}{2} |-6+1| = \frac{1}{2} |-5| = \frac{5}{2}$$

$$a_{31} = \frac{1}{2} |-3(3)+1| = \frac{1}{2} |-9+1| = \frac{1}{2} |-8| = \frac{8}{2} = 4$$

$$a_{12} = \frac{1}{2} |-3(1)+2| = \frac{1}{2} |-3+2| = \frac{1}{2} |-1| = \frac{1}{2}$$

$$a_{22} = \frac{1}{2} |-3(2)+2| = \frac{1}{2} |-6+2| = \frac{1}{2} |-4| = \frac{4}{2} = 2$$

$$a_{32} = \frac{1}{2} |-3(3)+2| = \frac{1}{2} |-9+2| = \frac{1}{2} |-7| = \frac{7}{2}$$

$$a_{13} = \frac{1}{2} |-3(1)+3| = \frac{1}{2} |-3+3| = 0$$

$$a_{23} = \frac{1}{2} |-3(2)+3| = \frac{1}{2} |-6+3| = \frac{1}{2} |-3| = \frac{3}{2}$$

$$a_{33} = \frac{1}{2} |-3(3)+3| = \frac{1}{2} |-9+3| = \frac{1}{2} |-6| = \frac{6}{2} = 3$$

$$a_{14} = \frac{1}{2} |-3(1)+4| = \frac{1}{2} |-3+4| = \frac{1}{2} |1| = \frac{1}{2}$$

$$a_{24} = \frac{1}{2} |-3(2)+4| = \frac{1}{2} |-6+4| = \frac{1}{2} |-2| = \frac{2}{2} = 1$$

$$a_{34} = \frac{1}{2} |-3(3)+4| = \frac{1}{2} |-9+4| = \frac{1}{2} |-5| = \frac{5}{2}$$

Conclusion

Thus, the required matrix is

$$A = \begin{pmatrix} 1 & \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{5}{2} & 2 & \frac{3}{2} & 1 \\ 4 & \frac{7}{2} & 3 & \frac{5}{2} \end{pmatrix}$$

Substitute

(ii) $a_{ij} = 2i - j$; $i = 1, 2, 3$ $j = 1, 2, 3, 4$

$$a_{11} = 2(1) - 1 = 2 - 1 = 1$$

$$a_{21} = 2(2) - 1 = 4 - 1 = 3$$

$$a_{31} = 2(3) - 1 = 6 - 1 = 5$$

$$a_{12} = 2(1) - 2 = 2 - 2 = 0$$

$$a_{22} = 2(2) - 2 = 4 - 2 = 2$$

$$a_{32} = 2(3) - 2 = 6 - 2 = 4$$

$$a_{13} = 2(1) - 3 = 2 - 3 = -1$$

$$a_{23} = 2(2) - 3 = 4 - 3 = 1$$

$$a_{33} = 2(3) - 3 = 6 - 3 = 3$$

$$a_{14} = 2(1) - 4 = 2 - 4 = -2$$

$$a_{24} = 2(2) - 4 = 4 - 4 = 0$$

$$a_{34} = 2(3) - 4 = 6 - 4 = 2$$

Conclusion

Thus, the required matrix is

$$A = \begin{pmatrix} 1 & 0 & -1 & -2 \\ 3 & 2 & 1 & 0 \\ 5 & 4 & 3 & 2 \end{pmatrix}$$

Q

6

Solving for
unknowns by
matrix
equality

(())

Find the values of x, y and z from the following equations:

Part (i)

$$\begin{bmatrix} 4 & 3 \\ x & 5 \end{bmatrix} = \begin{bmatrix} y & z \\ 1 & 5 \end{bmatrix}$$

Part (ii)

$$\begin{bmatrix} x+y & 2 \\ 5+z & x y \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 5 & 8 \end{bmatrix}$$

Part (iii)

$$\begin{bmatrix} x+y+z \\ x+z \\ y+z \end{bmatrix} = \begin{bmatrix} 9 \\ 5 \\ 7 \end{bmatrix}$$

Given

$$(i) \begin{pmatrix} 4 & 3 \\ x & 5 \end{pmatrix} = \begin{pmatrix} y & z \\ 1 & 5 \end{pmatrix}$$

As the given matrices are equal, their corresponding elements are also equal.

Comparing the corresponding elements, we get:

Conclusion

$$x = 1, y = 4 \text{ and } z = 3$$

Substitute

$$(ii) \begin{pmatrix} x+y & 2 \\ 5+z & xy \end{pmatrix} = \begin{pmatrix} 6 & 2 \\ 5 & 8 \end{pmatrix}$$

As the given matrices are equal, their corresponding elements are also equal.

Comparing the corresponding elements, we get:

$$x+y = 6$$

$$xy = 8$$

$$5+z = 5$$

Hence,

$$\Rightarrow 5+z = 5$$

$$\Rightarrow z = 0$$

Key formula

$$\text{We know that } (a-b)^2 = (a+b)^2 - 4ab$$

Substitute

$$\Rightarrow (x-y)^2 = (6)^2 - 8 \times 4$$

$$\Rightarrow (x-y)^2 = 36 - 32$$

$$\Rightarrow (x-y)^2 = 4$$

$$\Rightarrow (x-y) = \pm 2$$

Equating $x-y = 2$ and $x+y = 6$, we get $x = 4, y = 2$

Similarly, equating $x-y = -2$ and $x+y = 6$, we get $x = 2, y = 4$

Conclusion

$$\text{Thus, } x = 4, y = 2, z = 0 \text{ or } x = 2, y = 4, z = 0$$

Substitute

$$(iii) \begin{pmatrix} x+y+z \\ x+z \\ y+z \end{pmatrix} = \begin{pmatrix} 9 \\ 5 \\ 7 \end{pmatrix}$$

As the given matrices are equal, their corresponding elements are also equal.

Comparing the corresponding elements, we get:

$$x+y+z = 9$$

$$x+z = 5$$

$$y+z = 7$$

From (1) and (2), we have

$$\Rightarrow y+5 = 9$$

$$\Rightarrow y = 4$$

From (3), we have

$$\Rightarrow 4+z = 7$$

$$\Rightarrow z = 3$$

Therefore,

$$\Rightarrow x+z = 5$$

$$\Rightarrow x+3 = 5$$

$$\Rightarrow x = 2$$

Conclusion

Thus, $x = 2$, $y = 4$, $z = 3$

Q

7

Solving for
unknowns by
matrix
equality



Find the value of a, b, c and d from the equation:

$$\begin{bmatrix} a-b & 2a+c \\ 2a-b & 3c+d \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 0 & 13 \end{bmatrix}$$

Given

$$\begin{pmatrix} a-b & 2a+c \\ 2a-b & 3c+d \end{pmatrix} = \begin{pmatrix} -1 & 5 \\ 0 & 13 \end{pmatrix}$$

Substitute

As the two matrices are equal, their corresponding elements are also equal.

Comparing the corresponding elements, we get:

$$a-b = -1$$

$$2a-b = 0$$

$$2a+c = 5$$

$$3c+d = 13$$

From (2),

$$b = 2a$$

Putting this value in (1),

$$\Rightarrow a-2a = -1$$

$$\Rightarrow a = 1$$

Hence,

$$\Rightarrow b = 2$$

Putting $a = 1$ in (3),

$$\Rightarrow 2(1)+c = 5$$

$$\Rightarrow c = 3$$

Putting $c = 3$ in (4),

$$\Rightarrow 3(3)+d = 13$$

$$\Rightarrow d = 4$$

Conclusion

Thus, $a = 1$, $b = 2$, $c = 3$ and $d = 4$.

Q

8

MCQ -
condition for
a square
matrix



$A = [a_{ij}]_{m \times n}$ is a square matrix, if (A) $m < n$ (B) $m > n$ (C) $m = n$ (D) None of these

Given

A matrix is called a square matrix if the number of rows is equal to the number of columns.

Conclusion

Therefore, $A = [a_{ij}]_{m \times n}$ is a square matrix, if $m = n$.

Thus, the correct option is C.

Q

9

MCQ - values
making two
matrices
equal



Which of the given values of x and y make the following pair of matrices equal

$$\begin{bmatrix} 3x+7 & 5 \\ y+1 & 2-3x \end{bmatrix}, \begin{bmatrix} 0 & y-2 \\ 8 & 4 \end{bmatrix} \quad \text{(A) } x = \frac{-1}{3}, y = 7 \quad \text{(B) Not possible to find} \quad \text{(C) } y = 7, x = \frac{-2}{3}$$

$$\text{(D) } x = \frac{-1}{3}, y = \frac{-2}{3}$$

Given

The given matrices are $\begin{bmatrix} 3x+7 & 5 \\ y+1 & 2-3x \end{bmatrix}$ and $\begin{bmatrix} 0 & y-2 \\ 8 & 4 \end{bmatrix}$

Substitute

Equating the corresponding elements, we get:

$$3x+7 = 0 \Rightarrow x = \frac{-7}{3}$$

$$y-2 = 5 \Rightarrow y = 7$$

$$y+1 = 8 \Rightarrow y = 7$$

$$2-3x = 4 \Rightarrow x = \frac{-2}{3}$$

Conclusion

On comparing the corresponding elements of the two matrices, we get two different values of x .

This is not possible.

Hence, it is not possible to find the values of x and y for which the given matrices are equal.

Thus, the correct option is B.

Q

10

MCQ - count
of 0/1
matrices of
order 3×3



The number of all possible matrices of order 3×3 with each entry 0 or 1 is: (A) 27 (B) 18 (C) 81 (D) 512

Given

The given matrix of the order 3×3 has 9 elements and each of these elements can be either 0 or 1.

Substitute

Now, each of the 9 elements can be filled in two possible ways.

Hence, by the multiplication principle, the required number of possible matrices is $2^9 = 512$.

Conclusion

Thus, the correct option is D.

Q**11**

**Matrix
addition,
subtraction
and
multiplication**



Let $A = \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix}$, $C = \begin{bmatrix} -2 & 5 \\ 3 & 4 \end{bmatrix}$ Find each of the following:

Part (i)

A+B

Part (ii)

A - B

Part (iii)

3A - C

Part (iv)

AB

Part (v)

BA**Conclusion**

(i) A+B

$$\begin{aligned} &\Rightarrow \begin{pmatrix} 2 & 4 \\ 3 & 2 \end{pmatrix} + \begin{pmatrix} 1 & 3 \\ -2 & 5 \end{pmatrix} \\ &\Rightarrow \begin{pmatrix} 2+1 & 4+3 \\ 3-2 & 2+5 \end{pmatrix} \\ &\Rightarrow \begin{pmatrix} 3 & 7 \\ 1 & 7 \end{pmatrix} \end{aligned}$$

(ii) A-B

$$\begin{aligned} &\Rightarrow \begin{pmatrix} 2 & 4 \\ 3 & 2 \end{pmatrix} - \begin{pmatrix} 1 & 3 \\ -2 & 5 \end{pmatrix} \\ &\Rightarrow \begin{pmatrix} 2-1 & 4-3 \\ 3+2 & 2-5 \end{pmatrix} \\ &\Rightarrow \begin{pmatrix} 1 & 1 \\ 5 & -3 \end{pmatrix} \end{aligned}$$

(iii) 3 A-C

$$\begin{aligned} &\Rightarrow 3 \begin{pmatrix} 2 & 4 \\ 3 & 2 \end{pmatrix} - \begin{pmatrix} -2 & 5 \\ 3 & 4 \end{pmatrix} \\ &\Rightarrow \begin{pmatrix} 3 \times 2 & 3 \times 4 \\ 3 \times 3 & 3 \times 2 \end{pmatrix} - \begin{pmatrix} -2 & 5 \\ 3 & 4 \end{pmatrix} \\ &\Rightarrow \begin{pmatrix} 6+2 & 12-5 \\ 9-3 & 6-4 \end{pmatrix} \\ &\Rightarrow \begin{pmatrix} 8 & 7 \\ 6 & 2 \end{pmatrix} \end{aligned}$$

(iv) A B

$$\begin{aligned} &\Rightarrow \begin{pmatrix} 2 & 4 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ -2 & 5 \end{pmatrix} \\ &\Rightarrow \begin{pmatrix} 2(1)+4(-2) & 2(3)+4(5) \\ 3(1)+2(-2) & 3(3)+2(5) \end{pmatrix} \\ &\Rightarrow \begin{pmatrix} 2-8 & 6+20 \\ 3-4 & 9+10 \end{pmatrix} \\ &\Rightarrow \begin{pmatrix} -6 & 26 \\ -1 & 19 \end{pmatrix} \end{aligned}$$

(v) B A

$$\begin{aligned}
&\Rightarrow \begin{pmatrix} 1 & 3 \\ -2 & 5 \end{pmatrix} \begin{pmatrix} 2 & 4 \\ 3 & 2 \end{pmatrix} \\
&\Rightarrow \begin{pmatrix} 1(2)+3(3) & 1(4)+3(2) \\ -2(2)+5(3) & -2(4)+5(2) \end{pmatrix} \\
&\Rightarrow \begin{pmatrix} 2+9 & 4+6 \\ -4+15 & -8+10 \end{pmatrix} \\
&\Rightarrow \begin{pmatrix} 11 & 10 \\ 11 & 2 \end{pmatrix}
\end{aligned}$$

Q

12

Computing
sums of
matrices



Compute the following:

Part (i)

$$\begin{bmatrix} a & b \\ -b & a \end{bmatrix} + \begin{bmatrix} a & b \\ b & a \end{bmatrix}$$

Part (ii)

$$\begin{bmatrix} a^2+b^2 & b^2+c^2 \\ a^2+c^2 & a^2+b^2 \end{bmatrix} + \begin{bmatrix} 2ab & 2bc \\ -2ac & -2ab \end{bmatrix}$$

Part (iii)

$$\begin{bmatrix} -1 & 4 & -6 \\ 8 & 5 & 16 \\ 2 & 8 & 5 \end{bmatrix} + \begin{bmatrix} 12 & 7 & 6 \\ 8 & 0 & 5 \\ 3 & 2 & 4 \end{bmatrix}$$

Part (iv)

$$\begin{bmatrix} \cos^2 x & \sin^2 x \\ \sin^2 x & \cos^2 x \end{bmatrix} + \begin{bmatrix} \sin^2 x & \cos^2 x \\ \cos^2 x & \sin^2 x \end{bmatrix}$$

Conclusion

(i)

$$\begin{aligned}
&\begin{pmatrix} a & b \\ -b & a \end{pmatrix} + \begin{pmatrix} a & b \\ b & a \end{pmatrix} \\
&\Rightarrow \begin{pmatrix} a+a & b+b \\ -b+b & a+a \end{pmatrix} \\
&\Rightarrow \begin{pmatrix} 2a & 2b \\ 0 & 2a \end{pmatrix}
\end{aligned}$$

(ii)

$$\begin{aligned}
&\begin{pmatrix} a^2+b^2 & b^2+c^2 \\ a^2+c^2 & a^2+b^2 \end{pmatrix} + \begin{pmatrix} 2ab & 2bc \\ -2ac & -2ab \end{pmatrix} \\
&\Rightarrow \begin{pmatrix} a^2+b^2+2ab & b^2+c^2+2bc \\ a^2+c^2-2ac & a^2+b^2-2ab \end{pmatrix} \\
&\Rightarrow \begin{pmatrix} (a+b)^2 & (b+c)^2 \\ (a-c)^2 & (a-b)^2 \end{pmatrix}
\end{aligned}$$

(iii)

$$\begin{aligned}
&\begin{pmatrix} -1 & 4 & -6 \\ 8 & 5 & 16 \\ 2 & 8 & 5 \end{pmatrix} + \begin{pmatrix} 12 & 7 & 6 \\ 8 & 0 & 5 \\ 3 & 2 & 4 \end{pmatrix} \\
&\Rightarrow \begin{pmatrix} -1+12 & 4+7 & -6+6 \\ 8+8 & 5+0 & 16+5 \\ 2+3 & 8+2 & 5+4 \end{pmatrix} \\
&\Rightarrow \begin{pmatrix} 11 & 11 & 0 \\ 16 & 5 & 21 \\ 5 & 10 & 9 \end{pmatrix}
\end{aligned}$$

(iv)

$$\begin{aligned}
 & \begin{pmatrix} \cos^2 x & \sin^2 x \\ \sin^2 x & \cos^2 x \end{pmatrix} + \begin{pmatrix} \sin^2 x & \cos^2 x \\ \cos^2 x & \sin^2 x \end{pmatrix} \\
 & \Rightarrow \begin{pmatrix} \cos^2 x + \sin^2 x & \sin^2 x + \cos^2 x \\ \sin^2 x + \cos^2 x & \cos^2 x + \sin^2 x \end{pmatrix} \\
 & \Rightarrow \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}
 \end{aligned}$$

Q

13

Computing
matrix
products



Compute the indicated products.

Part (i)

$$\begin{bmatrix} a & b \\ -b & a \end{bmatrix} \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

Part (ii)

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 2 & 3 & 4 \end{bmatrix}$$

Part (iii)

$$\begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$$

Part (iv)

$$\begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & -3 & 5 \\ 0 & 2 & 4 \\ 3 & 0 & 5 \end{bmatrix}$$

Part (v)

$$\begin{bmatrix} 2 & 1 \\ 3 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ -1 & 2 & 1 \end{bmatrix}$$

Part (vi)

$$\begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & -3 \\ 1 & 0 \\ 3 & 1 \end{bmatrix}$$

Conclusion

(i)

$$\begin{aligned}
 & \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \\
 & \Rightarrow \begin{pmatrix} a(a)+b(b) & a(-b)+b(a) \\ -b(a)+a(b) & -b(-b)+a(a) \end{pmatrix} \\
 & \Rightarrow \begin{pmatrix} a^2+b^2 & -a b+a b \\ -a b+a b & b^2+a^2 \end{pmatrix} \\
 & \Rightarrow \begin{pmatrix} a^2+b^2 & 0 \\ 0 & a^2+b^2 \end{pmatrix}
 \end{aligned}$$

(ii)

$$\begin{aligned}
 & \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \begin{bmatrix} 2 & 3 & 4 \end{bmatrix} \\
 & \Rightarrow \begin{pmatrix} 1(2) & 1(3) & 1(4) \\ 2(2) & 2(3) & 2(4) \\ 3(2) & 3(3) & 3(4) \end{pmatrix} \\
 & \Rightarrow \begin{pmatrix} 2 & 3 & 4 \\ 4 & 6 & 8 \\ 6 & 9 & 12 \end{pmatrix}
 \end{aligned}$$

(iii)

$$\begin{aligned}
& \begin{pmatrix} 1 & -2 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \\
& \Rightarrow \begin{pmatrix} 1(1)-2(2) & 1(2)-2(3) & 1(3)-2(1) \\ 2(1)+3(2) & 2(2)+3(3) & 2(3)+3(1) \end{pmatrix} \\
& \Rightarrow \begin{pmatrix} -3 & -4 & 1 \\ 8 & 13 & 9 \end{pmatrix}
\end{aligned}$$

(iv)

$$\begin{aligned}
& \begin{pmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 1 & -3 & 5 \\ 0 & 2 & 4 \\ 3 & 0 & 5 \end{pmatrix} \\
& \Rightarrow \begin{pmatrix} 2(1)+3(0)+4(3) & 2(-3)+3(2)+4(0) & 2(5)+3(4)+4(5) \\ 3(1)+4(0)+5(3) & 3(-3)+4(2)+5(0) & 3(5)+4(4)+5(5) \\ 4(1)+5(0)+6(3) & 4(-3)+5(2)+6(0) & 4(5)+5(4)+6(5) \end{pmatrix} \\
& \Rightarrow \begin{pmatrix} 14 & 0 & 42 \\ 18 & -1 & 56 \\ 22 & -2 & 70 \end{pmatrix}
\end{aligned}$$

(v)

$$\begin{aligned}
& \begin{pmatrix} 2 & 1 \\ 3 & 2 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ -1 & 2 & 1 \end{pmatrix} \\
& \Rightarrow \begin{pmatrix} 2(1)+1(-1) & 2(0)+1(2) & 2(1)+1(1) \\ 3(1)+2(-1) & 3(0)+2(2) & 3(1)+2(1) \\ -1(1)+1(-1) & -1(0)+1(2) & -1(1)+1(1) \end{pmatrix} \\
& \Rightarrow \begin{pmatrix} 1 & 2 & 3 \\ 1 & 4 & 5 \\ -2 & 2 & 0 \end{pmatrix}
\end{aligned}$$

(vi)

$$\begin{aligned}
& \begin{pmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{pmatrix} \begin{pmatrix} 2 & -3 \\ 1 & 0 \\ 3 & 1 \end{pmatrix} \\
& \Rightarrow \begin{pmatrix} 3(2)-1(1)+3(3) & 3(-3)-1(0)+3(1) \\ -1(2)+0(1)+2(3) & -1(-3)+0(0)+2(1) \end{pmatrix} \\
& \Rightarrow \begin{pmatrix} 14 & -6 \\ 4 & 5 \end{pmatrix}
\end{aligned}$$

Q

14

Verifying
matrix
addition
associativity

(())

If $A = \begin{bmatrix} 1 & 2 & -3 \\ 5 & 0 & 2 \\ 1 & -1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 3 & -1 & 2 \\ 4 & 2 & 5 \\ 2 & 0 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} 4 & 1 & 2 \\ 0 & 3 & 2 \\ 1 & -2 & 3 \end{bmatrix}$, then compute $(A+B)$ and $(B-C)$. Also, verify that $A+(B-C) = (A+B)-C$.

Given

$$\begin{aligned}
(A+B) &= \begin{pmatrix} 1 & 2 & -3 \\ 5 & 0 & 2 \\ 1 & -1 & 1 \end{pmatrix} + \begin{pmatrix} 3 & -1 & 2 \\ 4 & 2 & 5 \\ 2 & 0 & 3 \end{pmatrix} \\
&= \begin{pmatrix} 4 & 1 & -1 \\ 9 & 2 & 7 \\ 3 & -1 & 4 \end{pmatrix} \\
(B-C) &= \begin{pmatrix} 3 & -1 & 2 \\ 4 & 2 & 5 \\ 2 & 0 & 3 \end{pmatrix} - \begin{pmatrix} 4 & 1 & 2 \\ 0 & 3 & 2 \\ 1 & -2 & 3 \end{pmatrix} \\
&= \begin{pmatrix} -1 & -2 & 0 \\ 4 & -1 & 3 \\ 1 & 2 & 0 \end{pmatrix}
\end{aligned}$$

Now,

$$\begin{aligned} A+(B-C) &= \begin{pmatrix} 1 & 2 & -3 \\ 5 & 0 & 2 \\ 1 & -1 & 1 \end{pmatrix} + \begin{pmatrix} -1 & -2 & 0 \\ 4 & -1 & 3 \\ 1 & 2 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & -3 \\ 9 & -1 & 5 \\ 2 & 1 & 1 \end{pmatrix} \\ (A+B)-C &= \begin{pmatrix} 4 & 1 & -1 \\ 9 & 2 & 7 \\ 3 & -1 & 4 \end{pmatrix} - \begin{pmatrix} 4 & 1 & 2 \\ 0 & 3 & 2 \\ 1 & -2 & 3 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & -3 \\ 9 & -1 & 5 \\ 2 & 1 & 1 \end{pmatrix} \end{aligned}$$

Conclusion

Hence, $A+(B-C) = (A+B)-C$.

Q

15

Computing
 $3A - 5B$



If $A = \begin{bmatrix} \frac{2}{3} & 1 & \frac{5}{3} \\ \frac{1}{3} & \frac{2}{3} & \frac{4}{3} \\ \frac{7}{3} & 2 & \frac{2}{3} \end{bmatrix}$ and $B = \begin{bmatrix} \frac{2}{5} & \frac{3}{5} & 1 \\ \frac{1}{5} & \frac{2}{5} & \frac{4}{5} \\ \frac{7}{5} & \frac{6}{5} & \frac{2}{5} \end{bmatrix}$, then compute $3A - 5B$.

Conclusion

$$\begin{aligned} 3A - 5B &= 3 \begin{pmatrix} \frac{2}{3} & 1 & \frac{5}{3} \\ \frac{1}{3} & \frac{2}{3} & \frac{4}{3} \\ \frac{7}{3} & 2 & \frac{2}{3} \end{pmatrix} - 5 \begin{pmatrix} \frac{2}{5} & \frac{3}{5} & 1 \\ \frac{1}{5} & \frac{2}{5} & \frac{4}{5} \\ \frac{7}{5} & \frac{6}{5} & \frac{2}{5} \end{pmatrix} \\ &= \begin{pmatrix} 2 & 3 & 5 \\ 1 & 2 & 4 \\ 7 & 6 & 2 \end{pmatrix} - \begin{pmatrix} 2 & 3 & 5 \\ 1 & 2 & 4 \\ 7 & 6 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \end{aligned}$$

Q

16

Simplifying a
trigonometric
matrix
expression



Simplify $\cos \theta \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} + \sin \theta \begin{bmatrix} \sin \theta & -\cos \theta \\ \cos \theta & \sin \theta \end{bmatrix}$

Conclusion

$$\begin{aligned} &\cos \theta \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} + \sin \theta \begin{pmatrix} \sin \theta & -\cos \theta \\ \cos \theta & \sin \theta \end{pmatrix} \\ \Rightarrow &\begin{pmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ -\sin \theta \cos \theta & \cos^2 \theta \end{pmatrix} + \begin{pmatrix} \sin^2 \theta & -\sin \theta \cos \theta \\ \sin \theta \cos \theta & \sin^2 \theta \end{pmatrix} \\ \Rightarrow &\begin{pmatrix} \cos^2 \theta + \sin^2 \theta & \sin \theta \cos \theta - \sin \theta \cos \theta \\ -\sin \theta \cos \theta + \sin \theta \cos \theta & \cos^2 \theta + \sin^2 \theta \end{pmatrix} \\ \Rightarrow &\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

Q**17**

Solving
simultaneous
matrix
equations for
X, Y



Find X and Y, if

Part (i)

$$\mathbf{X} + \mathbf{Y} = \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix} \text{ and } \mathbf{X} - \mathbf{Y} = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

Part (ii)

$$2\mathbf{X} + 3\mathbf{Y} = \begin{bmatrix} 2 & 3 \\ 4 & 0 \end{bmatrix} \text{ and } 3\mathbf{X} + 2\mathbf{Y} = \begin{bmatrix} 2 & -2 \\ -1 & 5 \end{bmatrix}$$

Given



(i)

$$\begin{aligned} \mathbf{X} + \mathbf{Y} &= \begin{pmatrix} 7 & 0 \\ 2 & 5 \end{pmatrix} \\ \mathbf{X} - \mathbf{Y} &= \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} \end{aligned}$$

Substitute

Adding equations (1) and (2),

$$\begin{aligned} 2\mathbf{X} &= \begin{pmatrix} 7 & 0 \\ 2 & 5 \end{pmatrix} + \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} \\ &= \begin{pmatrix} 10 & 0 \\ 2 & 8 \end{pmatrix} \\ \mathbf{X} &= \frac{1}{2} \begin{pmatrix} 10 & 0 \\ 2 & 8 \end{pmatrix} \\ &= \begin{pmatrix} 5 & 0 \\ 1 & 4 \end{pmatrix} \end{aligned}$$

Now,

$$\begin{aligned} \Rightarrow \mathbf{X} + \mathbf{Y} &= \begin{pmatrix} 7 & 0 \\ 2 & 5 \end{pmatrix} \\ \Rightarrow \mathbf{Y} &= \begin{pmatrix} 7 & 0 \\ 2 & 5 \end{pmatrix} - \begin{pmatrix} 5 & 0 \\ 1 & 4 \end{pmatrix} \\ \Rightarrow \mathbf{Y} &= \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} \end{aligned}$$

(ii)

$$\begin{aligned} 2\mathbf{X} + 3\mathbf{Y} &= \begin{pmatrix} 2 & 3 \\ 4 & 0 \end{pmatrix} \\ 3\mathbf{X} + 2\mathbf{Y} &= \begin{pmatrix} 2 & -2 \\ -1 & 5 \end{pmatrix} \end{aligned}$$

Multiplying equation (1) by 2,

$$\begin{aligned} 2(2\mathbf{X} + 3\mathbf{Y}) &= 2 \begin{pmatrix} 2 & 3 \\ 4 & 0 \end{pmatrix} \\ 4\mathbf{X} + 6\mathbf{Y} &= \begin{pmatrix} 4 & 6 \\ 8 & 0 \end{pmatrix} \end{aligned}$$

Multiplying equation (2) by 3,

$$\begin{aligned} 3(3\mathbf{X} + 2\mathbf{Y}) &= 3 \begin{pmatrix} 2 & -2 \\ -1 & 5 \end{pmatrix} \\ 9\mathbf{X} + 6\mathbf{Y} &= \begin{pmatrix} 6 & -6 \\ -3 & 15 \end{pmatrix} \end{aligned}$$

From (3) and (4),

$$\begin{aligned}
 (4X+6Y)-(9X+6Y) &= \begin{pmatrix} 4 & 6 \\ 8 & 0 \end{pmatrix} - \begin{pmatrix} 6 & -6 \\ -3 & 15 \end{pmatrix} \\
 -5X &= \begin{pmatrix} 4-6 & 6+6 \\ 8+3 & 0-15 \end{pmatrix} \\
 -5X &= \begin{pmatrix} -2 & 12 \\ 11 & -15 \end{pmatrix} \\
 X &= \frac{-1}{5} \begin{pmatrix} -2 & 12 \\ 11 & -15 \end{pmatrix} \\
 &= \begin{pmatrix} \frac{2}{5} & \frac{-12}{5} \\ \frac{-11}{5} & 3 \end{pmatrix}
 \end{aligned}$$

Conclusion

Now

$$\begin{aligned}
 \Rightarrow 2X+3Y &= \begin{pmatrix} 2 & 3 \\ 4 & 0 \end{pmatrix} \\
 \Rightarrow 2 \begin{pmatrix} \frac{2}{5} & \frac{-12}{5} \\ \frac{-11}{5} & 3 \end{pmatrix} + 3Y &= \begin{pmatrix} 2 & 3 \\ 4 & 0 \end{pmatrix} \\
 \Rightarrow \begin{pmatrix} \frac{4}{5} & \frac{-24}{5} \\ \frac{-22}{5} & 6 \end{pmatrix} + 3Y &= \begin{pmatrix} 2 & 3 \\ 4 & 0 \end{pmatrix} \\
 \Rightarrow 3Y &= \begin{pmatrix} 2 & 3 \\ 4 & 0 \end{pmatrix} - \begin{pmatrix} \frac{4}{5} & \frac{-24}{5} \\ \frac{-22}{5} & 6 \end{pmatrix} \\
 \Rightarrow 3Y &= \begin{pmatrix} \frac{6}{5} & \frac{39}{5} \\ \frac{42}{5} & -6 \end{pmatrix} \\
 \Rightarrow Y &= \frac{1}{3} \begin{pmatrix} \frac{6}{5} & \frac{39}{5} \\ \frac{42}{5} & -6 \end{pmatrix} \\
 \Rightarrow Y &= \begin{pmatrix} \frac{2}{5} & \frac{13}{5} \\ \frac{14}{5} & -2 \end{pmatrix}
 \end{aligned}$$

Q

18

Solving a
matrix
equation for
X

Find X, if $Y = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$ and $2X+Y = \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix}$

(())

Conclusion

$$\begin{aligned}
2X+Y &= \begin{pmatrix} 1 & 0 \\ -3 & 2 \end{pmatrix} \\
\Rightarrow 2X + \begin{pmatrix} 3 & 2 \\ 1 & 4 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ -3 & 2 \end{pmatrix} \\
\Rightarrow 2X &= \begin{pmatrix} 1 & 0 \\ -3 & 2 \end{pmatrix} - \begin{pmatrix} 3 & 2 \\ 1 & 4 \end{pmatrix} \\
\Rightarrow 2X &= \begin{pmatrix} -2 & -2 \\ -4 & -2 \end{pmatrix} \\
\Rightarrow X &= \frac{1}{2} \begin{pmatrix} -2 & -2 \\ -4 & -2 \end{pmatrix} \\
\Rightarrow X &= \begin{pmatrix} -1 & -1 \\ -2 & -1 \end{pmatrix}
\end{aligned}$$

Q**19**

Solving for x
and y in a
matrix
equation



Find x and y, if $2 \begin{bmatrix} 1 & 3 \\ 0 & x \end{bmatrix} + \begin{bmatrix} y & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 1 & 8 \end{bmatrix}$

Given

$$\begin{aligned}
\Rightarrow 2 \begin{pmatrix} 1 & 3 \\ 0 & x \end{pmatrix} + \begin{pmatrix} y & 0 \\ 1 & 2 \end{pmatrix} &= \begin{pmatrix} 5 & 6 \\ 1 & 8 \end{pmatrix} \\
\Rightarrow \begin{pmatrix} 2 & 6 \\ 0 & 2x \end{pmatrix} + \begin{pmatrix} y & 0 \\ 1 & 2 \end{pmatrix} &= \begin{pmatrix} 5 & 6 \\ 1 & 8 \end{pmatrix} \\
\Rightarrow \begin{pmatrix} 2+y & 6 \\ 1 & 2x+2 \end{pmatrix} &= \begin{pmatrix} 5 & 6 \\ 1 & 8 \end{pmatrix}
\end{aligned}$$

Comparing the corresponding elements of these two matrices,

$$2+y = 5$$

$$\Rightarrow y = 3$$

$$2x+2 = 8$$

$$\Rightarrow x = 3$$

Conclusion

Therefore, $x = 3$ and $y = 3$.

Q**20**

Solving for x,
y, z, t in a
matrix
equation



Solve the equation for x, y, z and t, if $2 \begin{bmatrix} x & z \\ y & t \end{bmatrix} + 3 \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} = 3 \begin{bmatrix} 3 & 5 \\ 4 & 6 \end{bmatrix}$

Given

$$\begin{aligned}
\Rightarrow 2 \begin{pmatrix} x & z \\ y & t \end{pmatrix} + 3 \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix} &= 3 \begin{pmatrix} 3 & 5 \\ 4 & 6 \end{pmatrix} \\
\Rightarrow \begin{pmatrix} 2x & 2z \\ 2y & 2t \end{pmatrix} + \begin{pmatrix} 3 & -3 \\ 0 & 6 \end{pmatrix} &= \begin{pmatrix} 9 & 15 \\ 12 & 18 \end{pmatrix} \\
\Rightarrow \begin{pmatrix} 2x+3 & 2z-3 \\ 2y & 2t+6 \end{pmatrix} &= \begin{pmatrix} 9 & 15 \\ 12 & 18 \end{pmatrix}
\end{aligned}$$

Comparing the corresponding elements of these two matrices,

$$2x+3 = 9$$

$$\Rightarrow 2x = 6$$

$$\Rightarrow x = 3$$

$$2y = 12$$

$$\Rightarrow y = 6$$

$$2z-3 = 15$$

$$\Rightarrow 2z = 18$$

$$\Rightarrow z = 9$$

$$\begin{aligned}
 2t+6 &= 18 \\
 \Rightarrow 2t &= 12 \\
 \Rightarrow t &= 6
 \end{aligned}$$

Conclusion

Therefore, $x = 3$, $y = 6$, $z = 9$ and $t = 6$.

Q

21

Solving for x and y using a vector equation



If $x \begin{bmatrix} 2 \\ 3 \end{bmatrix} + y \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$, find the values of x and y .

Given



$$\begin{aligned}
 \Rightarrow x \begin{pmatrix} 2 \\ 3 \end{pmatrix} + y \begin{pmatrix} -1 \\ 1 \end{pmatrix} &= \begin{pmatrix} 10 \\ 5 \end{pmatrix} \\
 \Rightarrow \begin{pmatrix} 2x \\ 3x \end{pmatrix} + \begin{pmatrix} -y \\ y \end{pmatrix} &= \begin{pmatrix} 10 \\ 5 \end{pmatrix} \\
 \Rightarrow \begin{pmatrix} 2x-y \\ 3x+y \end{pmatrix} &= \begin{pmatrix} 10 \\ 5 \end{pmatrix}
 \end{aligned}$$

Comparing the corresponding elements of these two matrices,

$$\begin{aligned}
 2x-y &= 10 \\
 3x+y &= 5
 \end{aligned}$$

By adding these two equations, we get

$$\begin{aligned}
 5x &= 15 \\
 \Rightarrow x &= 3
 \end{aligned}$$

Now, putting this value in (2)

$$\begin{aligned}
 \Rightarrow 3x+y &= 5 \\
 \Rightarrow y &= 5-3x \\
 \Rightarrow y &= 5-3(3) \\
 \Rightarrow y &= 5-9 \\
 \Rightarrow y &= -4
 \end{aligned}$$

Conclusion

Therefore, $x = 3$ and $y = 4$.

Q

22

Solving for x , y , z , w in a matrix equation

(())

Given $3 \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} x & 6 \\ -1 & 2w \end{bmatrix} + \begin{bmatrix} 4 & x+y \\ z+w & 3 \end{bmatrix}$, find the values of x , y , z and w .

Given



$$\begin{aligned}
 \Rightarrow 3 \begin{pmatrix} x & y \\ z & w \end{pmatrix} &= \begin{pmatrix} x & 6 \\ -1 & 2w \end{pmatrix} + \begin{pmatrix} 4 & x+y \\ z+w & 3 \end{pmatrix} \\
 \Rightarrow \begin{pmatrix} 3x & 3y \\ 3z & 3w \end{pmatrix} &= \begin{pmatrix} x+4 & 6+x+y \\ -1+z+w & 2w+3 \end{pmatrix}
 \end{aligned}$$

Substitute

Comparing the corresponding elements of these two matrices,

$$\begin{aligned}
 \Rightarrow 3x &= x+4 \\
 \Rightarrow 2x &= 4 \\
 \Rightarrow x &= 2 \\
 \Rightarrow 3y &= 6+x+y \\
 \Rightarrow 2y &= 6+x \\
 \Rightarrow 2y &= 6+2 \\
 \Rightarrow 2y &= 8 \\
 \Rightarrow y &= 4 \\
 \Rightarrow 3w &= -1+z+w+3 \\
 \Rightarrow 2w &= z+2
 \end{aligned}$$

$$\Rightarrow 3z = -1 + z + w$$

$$\Rightarrow 2z = w - 1$$

$$\Rightarrow 2z = 3 - 1$$

$$\Rightarrow 2z = -2$$

$$\Rightarrow z = 1$$

Conclusion

Therefore, $x = 2$, $y = 4$, $z = 1$ and $w = 3$

Q

23

**Matrix
function
identity**
 $F(x)F(y) = F(x+y)$



If $F(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$, show that $F(x)F(y) = F(x+y)$.

Given

It is given that

$$F(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Then,

$$F(y) = \begin{bmatrix} \cos y & -\sin y & 0 \\ \sin y & \cos y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now,

$$F(x+y) = \begin{bmatrix} \cos(x+y) & -\sin(x+y) & 0 \\ \sin(x+y) & \cos(x+y) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Substitute

$$\begin{aligned} F(x)F(y) &= \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos y & -\sin y & 0 \\ \sin y & \cos y & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \cos x \cos y - \sin x \sin y + 0 & -\cos x \sin y - \sin x \cos y + 0 & 0 \\ \sin x \cos y + \cos x \sin y + 0 & -\sin x \sin y + \cos x \cos y + 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \cos(x+y) & -\sin(x+y) & 0 \\ \sin(x+y) & \cos(x+y) & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= F(x+y) \end{aligned}$$

Conclusion

Therefore, $F(x)F(y) = F(x+y)$

Q

24

**Showing
matrix
multiplication
is non-
commutative**



Show that

Part (i)

$$\begin{bmatrix} 5 & -1 \\ 6 & 7 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} \neq \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & -1 \\ 6 & 7 \end{bmatrix}$$

Part (ii)

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 2 & 3 & 4 \end{bmatrix} \neq \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

Given

$$(i) \begin{bmatrix} 5 & -1 \\ 6 & 7 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} \neq \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & -1 \\ 6 & 7 \end{bmatrix}$$



$$\begin{pmatrix} 5 & -1 \\ 6 & 7 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 5(2)-1(3) & 5(1)-1(4) \\ 6(2)+7(3) & 6(1)+7(4) \end{pmatrix}$$

$$= \begin{pmatrix} 10-3 & 5-4 \\ 12+21 & 6+28 \end{pmatrix}$$

$$= \begin{pmatrix} 7 & 1 \\ 33 & 34 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & -1 \\ 6 & 7 \end{pmatrix} = \begin{pmatrix} 2(5)+1(6) & 2(-1)+1(7) \\ 3(5)+4(6) & 3(-1)+4(7) \end{pmatrix}$$

$$= \begin{pmatrix} 10+6 & -2+7 \\ 15+24 & -3+28 \end{pmatrix}$$

$$= \begin{pmatrix} 16 & 5 \\ 39 & 25 \end{pmatrix}$$

$$\text{Thus, } \begin{pmatrix} 5 & -1 \\ 6 & 7 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix} \neq \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 & -1 \\ 6 & 7 \end{pmatrix}$$

Conclusion

$$(ii) \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 2 & 3 & 4 \end{pmatrix} \neq \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 2 & 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 2 & 3 & 4 \end{pmatrix} = \begin{pmatrix} 1(-1)+2(0)+3(2) & 1(1)+2(-1)+3(3) & 1(0)+2(1)+3(4) \\ 0(-1)+1(0)+0(2) & 0(1)+1(-1)+0(3) & 0(0)+1(1)+0(4) \\ 1(-1)+1(0)+0(2) & 1(1)+1(-1)+0(3) & 1(0)+1(1)+0(4) \end{pmatrix}$$

$$= \begin{pmatrix} 5 & 8 & 14 \\ 0 & -1 & 1 \\ -1 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 2 & 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix} = \begin{pmatrix} -1(1)+1(0)+0(1) & -1(2)+1(1)+0(1) & -1(3)+1(0)+0(0) \\ 0(1)+(-1)(0)+1(1) & 0(2)+(-1)(1)+1(1) & 0(3)+(-1)(0)+1(0) \\ 2(1)+3(0)+4(1) & 2(2)+3(1)+4(1) & 2(3)+3(0)+4(0) \end{pmatrix}$$

$$= \begin{pmatrix} -1 & -1 & -3 \\ 1 & 0 & 0 \\ 6 & 11 & 6 \end{pmatrix}$$

$$\text{Thus, } \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 2 & 3 & 4 \end{pmatrix} \neq \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 2 & 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$$

Q

25

Evaluating a
matrix
polynomial



Find $A^2 - 5A + 6I$, if $A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$

Given



$$A^2 = A \cdot A$$

$$= \begin{pmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 2(2)+0(2)+1(1) & 2(0)+0(1)+1(-1) & 2(1)+0(3)+1(0) \\ 2(2)+1(2)+3(1) & 2(0)+1(1)+1(1) & 2(1)+1(3)+3(0) \\ 1(2)+(-1)(2)+0(1) & 1(0)+(-1)(1)+0(-1) & 1(1)+(-1)(3)+0(0) \end{pmatrix}$$

$$= \begin{pmatrix} 4+0+1 & 0+0-1 & 2+0+0 \\ 4+2+3 & 0+1-1 & 2+3+0 \\ 2-2+0 & 0-1+0 & 1-3+0 \end{pmatrix}$$

$$= \begin{pmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{pmatrix}$$

Conclusion

Therefore,

$$\begin{aligned}
 A^2 - 5A + 6I &= \begin{pmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{pmatrix} - 5 \begin{pmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{pmatrix} + 6 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{pmatrix} - \begin{pmatrix} 10 & 0 & 5 \\ 10 & 5 & 15 \\ 5 & -5 & 0 \end{pmatrix} + \begin{pmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{pmatrix} \\
 &= \begin{pmatrix} 5-10 & -1-0 & 2-5 \\ 9-10 & -2-5 & 5-15 \\ 0-5 & -1+5 & -2-0 \end{pmatrix} + \begin{pmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{pmatrix} \\
 &= \begin{pmatrix} -5 & -1 & -3 \\ -1 & -7 & -10 \\ -5 & 4 & -2 \end{pmatrix} + \begin{pmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{pmatrix} \\
 &= \begin{pmatrix} 1 & -1 & -3 \\ -1 & -1 & -10 \\ -5 & 4 & 4 \end{pmatrix}
 \end{aligned}$$

Q

26

Proving a
matrix
polynomial
identity

(())

If $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$, prove that $A^3 - 6A^2 + 7A + 2I = 0$

Given



$$\begin{aligned}
 A^2 &= A \cdot A \\
 &= \begin{pmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{pmatrix} \\
 &= \begin{pmatrix} 1+0+4 & 0+0+0 & 2+0+6 \\ 0+0+2 & 0+4+0 & 0+2+3 \\ 2+0+6 & 0+0+0 & 4+0+9 \end{pmatrix} \\
 &= \begin{pmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{pmatrix}
 \end{aligned}$$

Substitute

Now,

$$\begin{aligned}
 A^3 &= A^2 \cdot A \\
 &= \begin{pmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{pmatrix} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{pmatrix} \\
 &= \begin{pmatrix} 5+0+16 & 0+0+0 & 10+0+24 \\ 2+0+10 & 0+8+0 & 4+4+15 \\ 8+0+26 & 0+0+0 & 16+0+39 \end{pmatrix} \\
 &= \begin{pmatrix} 21 & 0 & 34 \\ 12 & 8 & 23 \\ 34 & 0 & 55 \end{pmatrix}
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 A^3 - 6A^2 + 7A + 2I &= \begin{pmatrix} 21 & 0 & 34 \\ 12 & 8 & 23 \\ 34 & 0 & 55 \end{pmatrix} - 6 \begin{pmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{pmatrix} + 7 \begin{pmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{pmatrix} + 2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} 21 & 0 & 34 \\ 12 & 8 & 23 \\ 34 & 0 & 55 \end{pmatrix} - \begin{pmatrix} 30 & 0 & 48 \\ 12 & 24 & 30 \\ 48 & 0 & 78 \end{pmatrix} + \begin{pmatrix} 7 & 0 & 14 \\ 0 & 14 & 7 \\ 14 & 0 & 21 \end{pmatrix} + \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \\
 &= \begin{pmatrix} 21+7+2 & 0+0+0 & 34+14+0 \\ 12+0+0 & 8+14+2 & 23+7+0 \\ 34+14+0 & 0+0+0 & 55+21+2 \end{pmatrix} - \begin{pmatrix} 30 & 0 & 48 \\ 12 & 24 & 30 \\ 48 & 0 & 78 \end{pmatrix}
 \end{aligned}$$

$$= \begin{pmatrix} 30 & 0 & 48 \\ 12 & 24 & 30 \\ 48 & 0 & 78 \end{pmatrix} - \begin{pmatrix} 30 & 0 & 48 \\ 12 & 24 & 30 \\ 48 & 0 & 78 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 0$$

Conclusion

Hence, $A^3 - 6A^2 + 7A + 2I = 0$.

Q

27

Finding k in
a matrix
equation
 $A^2 = kA - 2I$



If $A = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, find k so that $A^2 = kA - 2I$

Given



$$A^2 = A \cdot A$$

$$= \begin{pmatrix} 3 & -2 \\ 4 & -2 \end{pmatrix} \begin{pmatrix} 3 & -2 \\ 4 & -2 \end{pmatrix}$$

$$= \begin{pmatrix} 3(3)+(-2)(4) & 3(-2)+(-2)(-2) \\ 4(3)+(-2)(4) & 4(-2)+(-2)(-2) \end{pmatrix}$$

$$= \begin{pmatrix} 1 & -2 \\ 4 & -4 \end{pmatrix}$$

Substitute

Now,

$$\Rightarrow A^2 = kA - 2I$$

$$\Rightarrow \begin{pmatrix} 1 & -2 \\ 4 & -4 \end{pmatrix} = k \begin{pmatrix} 3 & -2 \\ 4 & -2 \end{pmatrix} - 2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & -2 \\ 4 & -4 \end{pmatrix} = \begin{pmatrix} 3k & -2k \\ 4k & -2k \end{pmatrix} - \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & -2 \\ 4 & -4 \end{pmatrix} = \begin{pmatrix} 3k-2 & -2k \\ 4k & -2k-2 \end{pmatrix}$$

Comparing the corresponding elements, we have:

$$3k - 2 = 1$$

$$\Rightarrow 3k = 3$$

$$\Rightarrow k = 1$$

Conclusion

Therefore, the value of k = 1.

Q

28

Matrix
identity with
 $\tan(\alpha/2)$



If $A = \begin{bmatrix} 0 & -\tan \frac{\alpha}{2} \\ \tan \frac{\alpha}{2} & 0 \end{bmatrix}$ and I is the identity matrix of order 2, show that

$$I + A = (I - A) \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$$

Given

$$LHS = I + A$$



$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & -\tan \frac{\alpha}{2} \\ \tan \frac{\alpha}{2} & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & -\tan \frac{\alpha}{2} \\ \tan \frac{\alpha}{2} & 1 \end{pmatrix}$$

Substitute

$$RHS = (I - A) \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 0 & -\tan \frac{\alpha}{2} \\ \tan \frac{\alpha}{2} & 0 \end{pmatrix} \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$$

$$= \begin{pmatrix} 1 & \tan \frac{\alpha}{2} \\ -\tan \frac{\alpha}{2} & 1 \end{pmatrix} \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$$

$$= \begin{pmatrix} \cos \alpha + \sin \alpha \tan \frac{\alpha}{2} & -\sin \alpha + \cos \alpha \tan \frac{\alpha}{2} \\ -\cos \alpha \tan \frac{\alpha}{2} + \sin \alpha & \sin \alpha \tan \frac{\alpha}{2} + \cos \alpha \end{pmatrix}$$

$$= \begin{pmatrix} 1 - 2 \sin^2 \frac{\alpha}{2} + 2 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2} \tan \frac{\alpha}{2} & -2 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2} + (2 \cos^2 \frac{\alpha}{2} - 1) \tan \frac{\alpha}{2} \\ -(2 \cos^2 \frac{\alpha}{2} - 1) \tan \frac{\alpha}{2} + 2 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2} & \cos \frac{\alpha}{2} \sin \frac{\alpha}{2} + 1 - 2 \sin^2 \frac{\alpha}{2} \end{pmatrix}$$

$$= \begin{pmatrix} 1 - 2 \sin^2 \frac{\alpha}{2} + 2 \sin^2 \frac{\alpha}{2} & -2 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2} + 2 \sin^2 \frac{\alpha}{2} \cos^2 \frac{\alpha}{2} \\ -2 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2} + \tan \frac{\alpha}{2} + 2 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2} & \cos^2 \frac{\alpha}{2} + 1 - 2 \sin^2 \frac{\alpha}{2} \end{pmatrix}$$

$$= \begin{pmatrix} 1 & -\tan \frac{\alpha}{2} \\ \tan \frac{\alpha}{2} & 1 \end{pmatrix}$$

Conclusion

Thus, from ⁽¹⁾ and (2), we get

$$I + A = (I - A) \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$$

Q

29

Matrix
multiplication
- trust fund
investment



A trust fund has ₹ 30,000 that must be invested in two different types of bonds. The first bond pays 5 % interest per year, and the second bond pays 7 % interest per year. Using matrix multiplication, determine how to divide ₹ 30,000 among the two types of bonds. If the trust fund must obtain an annual total interest of:

Part (a)

₹ 1800

Part (b)

₹2000

Given

(i) Let ₹ x be invested in the first bond.

Then the sum invested in the second bond will be ₹ (30000-x).

The first bond pays 5 % interest per year, and the second bond pays 7 % interest per year.

Key formula

To obtain an annual total interest of ₹1800, we have:

$$[x(30000-x)] \begin{bmatrix} \frac{5}{100} \\ \frac{7}{100} \end{bmatrix} = 1800 \quad \left[\text{S.I for 1 year} = \frac{\text{Principal} \times \text{Rate}}{100} \right]$$
$$\Rightarrow \frac{5x}{100} + \frac{7(30000-x)}{100} = 1800$$
$$\Rightarrow 5x + 210000 - 7x = 180000$$
$$\Rightarrow 210000 - 2x = 180000$$
$$\Rightarrow 2x = 210000 - 180000$$
$$\Rightarrow 2x = 30000$$
$$\Rightarrow x = 15000$$

Conclusion

Thus, to obtain an annual total interest of ₹1800, the trust fund should invest ₹15000 in the first bond and the remaining ₹15000 in the second bond.

Given

(ii) Let ₹ x be invested in the first bond.

Then the sum invested in the second bond will be ₹ (30000-x).

Key formula

To obtain an annual total interest of ₹2000, we have:

$$[x(30000-x)] \begin{bmatrix} \frac{5}{100} \\ \frac{7}{100} \end{bmatrix} = 2000$$
$$\Rightarrow \frac{5x}{100} + \frac{7(30000-x)}{100} = 2000$$
$$\Rightarrow 5x + 210000 - 7x = 200000$$
$$\Rightarrow 210000 - 2x = 200000$$
$$\Rightarrow 2x = 210000 - 200000$$
$$\Rightarrow 2x = 10000$$
$$\Rightarrow x = 5000$$

Conclusion

Thus, to obtain an annual total interest of ₹1800, the trust fund should invest ₹5000 in the first bond and the remaining ₹25000 in the second bond.

Q

30

**Matrix
multiplication
- bookshop
revenue**

()

The bookshop of a particular school has 10 dozen chemistry books, 8 dozen physics books, 10 dozen economics books. Their selling prices are ₹ 80, ₹ 60 and ₹ 40 each respectively. Find the total amount the bookshop will receive from selling all the books using matrix algebra.

Key formula

The total amount received from selling all these books can be written as a matrix product:



$$12 \begin{bmatrix} 10 & 8 & 10 \end{bmatrix} \begin{bmatrix} 80 \\ 60 \\ 40 \end{bmatrix} = 12[10(80)+8(60)+10(40)]$$

$$= 12(800+480+400)$$

$$= 12(1680)$$

$$= 20160$$

Conclusion

Thus, the bookshop will receive ₹ 20160 from the sale of all these books.

Q**31**

MCQ -
restriction
on matrix
order for a
sum



The restriction on n , k and p so that $PY+WY$ will be defined are: (A) $k = 3$, $p = n$
(B) k is arbitrary, $p = 2$ (C) p is arbitrary, $k = 3$ (D) $k = 2$, $p = 3$

Given

Matrices P and Y are of the orders $p \times k$ and $3 \times k$ respectively.

Therefore, matrix PY will be defined if $k = 3$.

Consequently, PY will be of the order $p \times k$.

Matrices W and Y are of the orders $n \times 3$ and $3 \times k$ respectively.

The number of columns in W equals the number of rows in Y .

So matrix WY is well defined and is of the order $n \times k$.

Matrices PY and WY can be added only when their orders are the same.

However, PY is of the order $p \times k$ and WY is of the order $n \times k$.

Therefore, we must have $p = n$.

Thus, $k = 3$ and $p = n$ are the restrictions on n , k and p so that $PY+WY$ will be defined.

Conclusion

The correct option is A.

Q**32**

MCQ - order
of a matrix
combination



If $n = p$, then the order of the matrix $7X-5Z$ is: (A) $p \times 2$ (B) $2 \times n$ (C) $n \times 3$
(D) $p \times n$

Given

Matrix X is of the order $2 \times n$.

Therefore, matrix $7X$ is also of the same order.

Matrix Z is of the order $2 \times p$, i.e., $2 \times n$ [Since $n = p$]

Therefore, matrix $5Z$ is also of the same order.

Now, both the matrices $7X$ and $5Z$ are of the order $2 \times n$.

Thus, matrix $7X-5Z$ is well-defined and is of the order $2 \times n$.

Conclusion

The correct option is B.

Q**33**

Transpose of
given
matrices



Find the transpose of each of the following matrices:

Part (i)

$$\begin{bmatrix} 5 \\ 1 \\ 2 \\ -1 \end{bmatrix}$$

Part (ii)

$$\begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$$

Part (iii)

$$\begin{bmatrix} -1 & 5 & 6 \\ \sqrt{3} & 5 & 6 \\ 2 & 3 & -1 \end{bmatrix}$$

Conclusion

(i) Let

$$A = \begin{pmatrix} 5 \\ \frac{1}{2} \\ -1 \end{pmatrix}$$

$$\text{Then } A^T = \begin{pmatrix} 5 & \frac{1}{2} & -1 \end{pmatrix}$$

$$\text{(ii) Let } A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}$$

$$\text{Then } A^T = \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix}$$

(iii) Let

$$A^T = \begin{pmatrix} -1 & \sqrt{3} & 2 \\ 5 & 5 & 3 \\ 6 & 6 & -1 \end{pmatrix}$$

Q

34

Verifying
transpose of
sum and
difference

(())

If $A = \begin{bmatrix} -1 & 2 & 3 \\ 5 & 7 & 9 \\ -2 & 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -4 & 1 & -5 \\ 1 & 2 & 0 \\ 1 & 3 & 1 \end{bmatrix}$, then verify that

Part (i)

$$(A+B)' = A' + B'$$

Part (ii)

$$(A-B)' = A' - B'$$

Given

It is given that

$$A = \begin{pmatrix} -1 & 2 & 3 \\ 5 & 7 & 9 \\ -2 & 1 & 1 \end{pmatrix} \text{ and } B = \begin{pmatrix} -4 & 1 & -5 \\ 1 & 2 & 0 \\ 1 & 3 & 1 \end{pmatrix}$$

Hence, we have

$$A' = \begin{pmatrix} -1 & 5 & -2 \\ 2 & 7 & 1 \\ 3 & 9 & 1 \end{pmatrix} \text{ and } B' = \begin{pmatrix} -4 & 1 & 1 \\ 1 & 2 & 3 \\ -5 & 0 & 1 \end{pmatrix}$$

(i)

$$(A+B) = \begin{pmatrix} -1 & 2 & 3 \\ 5 & 7 & 9 \\ -2 & 1 & 1 \end{pmatrix} + \begin{pmatrix} -4 & 1 & -5 \\ 1 & 2 & 0 \\ 1 & 3 & 1 \end{pmatrix} = \begin{pmatrix} -5 & 3 & -2 \\ 6 & 9 & 9 \\ -1 & 4 & 2 \end{pmatrix}$$

Hence,

$$(A+B)' = \begin{pmatrix} -5 & 6 & -1 \\ 3 & 9 & 4 \\ -2 & 9 & 2 \end{pmatrix}$$

Now,

$$\begin{aligned} A' + B' &= \begin{pmatrix} -1 & 5 & -2 \\ 2 & 7 & 1 \\ 3 & 9 & 1 \end{pmatrix} + \begin{pmatrix} -4 & 1 & 1 \\ 1 & 2 & 3 \\ -5 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} -5 & 6 & -1 \\ 3 & 9 & 4 \\ -2 & 9 & 2 \end{pmatrix} \end{aligned}$$

Conclusion

Thus, $(A+B)' = A' + B'$.

Substitute

(ii)

$$(A-B) = \begin{pmatrix} -1 & 2 & 3 \\ 5 & 7 & 9 \\ -2 & 1 & 1 \end{pmatrix} - \begin{pmatrix} -4 & 1 & -5 \\ 1 & 2 & 0 \\ 1 & 3 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 1 & 8 \\ 4 & 5 & 9 \\ -3 & -2 & 0 \end{pmatrix}$$

Hence,

$$(A-B)' = \begin{pmatrix} 3 & 4 & -3 \\ 1 & 5 & -2 \\ 8 & 9 & 0 \end{pmatrix}$$

Now,

$$\begin{aligned} A' - B' &= \begin{pmatrix} -1 & 5 & -2 \\ 2 & 7 & 1 \\ 3 & 9 & 1 \end{pmatrix} - \begin{pmatrix} -4 & 1 & 1 \\ 1 & 2 & 3 \\ -5 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 3 & 4 & -3 \\ 1 & 5 & -2 \\ 8 & 9 & 0 \end{pmatrix} \end{aligned}$$

Conclusion

Thus, $(A-B)' = A' - B'$.

Q

35

Verifying
transpose of
sum and
difference



If $A' = \begin{bmatrix} 3 & 4 \\ -1 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 2 & 1 \\ 1 & 2 & 3 \end{bmatrix}$, then verify that

Part (i)

$$(A+B)' = A' + B'$$

Part (ii)

$$(A-B)' = A' - B'$$

Given

It is known that $A = (A')'$

Hence,

$$A = \begin{pmatrix} 3 & -1 & 0 \\ 4 & 2 & 1 \end{pmatrix} \text{ and } B' = \begin{pmatrix} -1 & 1 \\ 2 & 2 \\ 1 & 3 \end{pmatrix}$$

$$(i) \ A+B = \begin{pmatrix} 3 & -1 & 0 \\ 4 & 2 & 1 \end{pmatrix} + \begin{pmatrix} -1 & 2 & 1 \\ 1 & 2 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 1 \\ 5 & 4 & 4 \end{pmatrix}$$

Therefore,

$$(A+B)' = \begin{pmatrix} 2 & 5 \\ 1 & 4 \\ 1 & 4 \end{pmatrix}$$

Now,

$$A' + B' = \begin{pmatrix} 3 & 4 \\ -1 & 2 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} -1 & 1 \\ 2 & 2 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 5 \\ 1 & 4 \\ 1 & 4 \end{pmatrix}$$

Conclusion

Hence, $(A+B)' = A' + B'$.

Substitute

(ii)

$$A-B = \begin{pmatrix} 3 & -1 & 0 \\ 4 & 2 & 1 \end{pmatrix} - \begin{pmatrix} -1 & 2 & 1 \\ 1 & 2 & 3 \end{pmatrix} = \begin{pmatrix} 4 & -3 & -1 \\ 3 & 0 & -2 \end{pmatrix}$$

Therefore,

$$(A-B)' = \begin{pmatrix} 4 & 3 \\ -3 & 0 \\ -1 & -2 \end{pmatrix}$$

Now,

$$A' - B' = \begin{pmatrix} 3 & 4 \\ -1 & 2 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} -1 & 1 \\ 2 & 2 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 4 & 3 \\ -3 & 0 \\ -1 & -2 \end{pmatrix}$$

Conclusion

Hence, $(A-B)' = A' - B'$.

Q

36

Finding the
transpose of
(A+2B)



If $A' = \begin{bmatrix} -2 & 3 \\ 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 0 \\ 1 & 2 \end{bmatrix}$, then find $(A+2B)'$

Conclusion

It is known that $A = (A')'$.

Therefore,

$$A = \begin{pmatrix} -2 & 1 \\ 3 & 2 \end{pmatrix}$$

Now,

$$\begin{aligned} A+2B &= \begin{pmatrix} -2 & 1 \\ 3 & 2 \end{pmatrix} + 2 \begin{pmatrix} -1 & 0 \\ 1 & 2 \end{pmatrix} \\ &= \begin{pmatrix} -2 & 1 \\ 3 & 2 \end{pmatrix} + \begin{pmatrix} -2 & 0 \\ 2 & 4 \end{pmatrix} \\ &= \begin{pmatrix} -4 & 1 \\ 5 & 6 \end{pmatrix} \end{aligned}$$

Q

37

Verifying
(AB)' = B'A'



For the matrices A and B, verify that $(AB)' = B' A'$, where

Part (i)

$$A = \begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix}, B = \begin{bmatrix} -1 & 2 & 1 \end{bmatrix}$$

Part (ii)

$$A = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, B = \begin{bmatrix} 1 & 5 & 7 \end{bmatrix}$$

Given



(i)

$$A = \begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix} \text{ and } B = \begin{bmatrix} -1 & 2 & 1 \end{bmatrix}$$

Hence,

$$\begin{aligned} A B &= \begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix} \begin{bmatrix} -1 & 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -1 & 2 & 1 \\ 4 & -8 & -4 \\ -3 & 6 & 3 \end{bmatrix} \end{aligned}$$

Therefore,

$$(A B)' = \begin{bmatrix} -1 & 4 & -3 \\ 2 & -8 & 6 \\ 1 & -4 & 3 \end{bmatrix}$$

Now,

$$A' = \begin{bmatrix} 1 & -4 & 3 \end{bmatrix} \text{ and } B' = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$$

Hence,

$$\begin{aligned} B' A' &= \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & -4 & 3 \end{bmatrix} \\ &= \begin{bmatrix} -1 & 4 & -3 \\ 2 & -8 & 6 \\ 1 & -4 & 3 \end{bmatrix} \end{aligned}$$

Conclusion

Thus, $(A B)' = B' A'$

Substitute

(ii) It is given that $B = \begin{bmatrix} 1 & 5 & 7 \end{bmatrix}$.

Hence,

$$\begin{aligned} A B &= \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 5 & 7 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 1 & 5 & 7 \\ 2 & 10 & 14 \end{bmatrix} \end{aligned}$$

Therefore,

$$(A B)' = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 5 & 10 \\ 0 & 7 & 14 \end{bmatrix}$$

Now,

$$A' = \begin{bmatrix} 0 & 1 & 2 \end{bmatrix} \text{ and } B' = \begin{bmatrix} 1 \\ 5 \\ 7 \end{bmatrix}$$

Therefore,

$$B' A' = \begin{bmatrix} 1 \\ 5 \\ 7 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 2 \\ 0 & 5 & 10 \\ 0 & 7 & 14 \end{bmatrix}$$

Conclusion

Thus, $(A B)' = B' A'$.

Q

38

Verifying
 $A'A=I$ for a
rotation-type
matrix

(())

If (i) $A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$, then verify that $A' A = I$ (ii) If $A = \begin{bmatrix} \sin \alpha & \cos \alpha \\ -\cos \alpha & \sin \alpha \end{bmatrix}$, then verify that $A' A = I$

Given

(i)

It is given that $A = \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix}$

Therefore,

$$A' = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$$

Now,

$$A' A = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix}$$

$$= \begin{pmatrix} \cos \alpha \cos \alpha + (-\sin \alpha)(-\sin \alpha) & \sin \alpha \cos \alpha + (-\sin \alpha) \cos \alpha \\ \sin \alpha \cos \alpha + \cos \alpha(-\sin \alpha) & \sin \alpha \sin \alpha + \cos \alpha \cos \alpha \end{pmatrix}$$

$$= \begin{pmatrix} \cos^2 \alpha + \sin^2 \alpha & \sin \alpha \cos \alpha - \sin \alpha \cos \alpha \\ \sin \alpha \cos \alpha - \sin \alpha \cos \alpha & \sin^2 \alpha + \cos^2 \alpha \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= I$$

Conclusion

Thus, $A' A = I$

Substitute

(ii)

It is given that $A = \begin{pmatrix} \sin \alpha & \cos \alpha \\ -\cos \alpha & \sin \alpha \end{pmatrix}$

Therefore,

$$A' = \begin{pmatrix} \sin \alpha & -\cos \alpha \\ \cos \alpha & \sin \alpha \end{pmatrix}$$

Now,

$$A' A = \begin{pmatrix} \sin \alpha & -\cos \alpha \\ \cos \alpha & \sin \alpha \end{pmatrix} \begin{pmatrix} \sin \alpha & \cos \alpha \\ -\cos \alpha & \sin \alpha \end{pmatrix}$$

$$= \begin{pmatrix} \sin \alpha \sin \alpha + (-\cos \alpha)(-\cos \alpha) & \sin \alpha \cos \alpha + (-\cos \alpha) \sin \alpha \\ \sin \alpha \cos \alpha + \sin \alpha(-\cos \alpha) & \sin \alpha \sin \alpha + \cos \alpha \cos \alpha \end{pmatrix}$$

$$= \begin{pmatrix} \cos^2 \alpha + \sin^2 \alpha & \sin \alpha \cos \alpha - \sin \alpha \cos \alpha \\ \sin \alpha \cos \alpha - \sin \alpha \cos \alpha & \sin^2 \alpha + \cos^2 \alpha \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= I$$

Conclusion

Thus, $A' A = I$

Q

39

Showing a matrix is symmetric or skew symmetric



Part (i)

Show that the matrix $A = \begin{bmatrix} 1 & -1 & 5 \\ -1 & 2 & 1 \\ 5 & 1 & 3 \end{bmatrix}$ is a symmetric matrix.

Part (ii)

Show that the matrix $A = \begin{bmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}$ is a skew symmetric matrix.

Given

$$(i) A = \begin{bmatrix} 1 & -1 & 5 \\ -1 & 2 & 1 \\ 5 & 1 & 3 \end{bmatrix}$$

Now,

$$A' = \begin{bmatrix} 1 & -1 & 5 \\ -1 & 2 & 1 \\ 5 & 1 & 3 \end{bmatrix} \\ = A$$

Conclusion

Hence, A is a symmetric matrix.

Substitute

$$(ii) A = \begin{bmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}$$

$$A' = \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix} \\ = - \begin{bmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix} \\ = -A$$

Conclusion

Hence, A is a skew symmetric matrix.

Q

40

Verifying $A+A'$ symmetric, $A-A'$ skew symmetric



For the matrix $A = \begin{bmatrix} 1 & 5 \\ 6 & 7 \end{bmatrix}$, verify that

Part (i)

$(A+A')$ is a symmetric matrix

Part (ii)

$(A - A')$ is a skew symmetric matrix

Given

$$\text{It is given that } A = \begin{bmatrix} 1 & 5 \\ 6 & 7 \end{bmatrix}$$

$$\text{Hence, } A' = \begin{bmatrix} 1 & 6 \\ 5 & 7 \end{bmatrix}$$

$$(i) (A+A') = \begin{bmatrix} 1 & 5 \\ 6 & 7 \end{bmatrix} + \begin{bmatrix} 1 & 6 \\ 5 & 7 \end{bmatrix} = \begin{bmatrix} 2 & 11 \\ 11 & 14 \end{bmatrix}$$

Therefore,

$$(A+A')' = \begin{pmatrix} 2 & 11 \\ 11 & 14 \end{pmatrix}$$

$$= (A+A')$$

Conclusion

Thus, $(A+A')$ is a symmetric matrix.

Substitute

$$(ii) (A-A') = \begin{pmatrix} 1 & 5 \\ 6 & 7 \end{pmatrix} - \begin{pmatrix} 1 & 6 \\ 5 & 7 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$$

Therefore,

$$(A-A')' = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$= -\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

$$= -(A-A')$$

Conclusion

Thus, $(A-A')$ is a skew symmetric matrix.

Q

41

Finding
symmetric
and skew-
symmetric
parts



Find $\frac{1}{2} (A+A')$ and $\frac{1}{2} (A-A')$, when $A = \begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix}$

Conclusion

It is given that

$$A = \begin{pmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{pmatrix}$$

Hence,

$$A' = \begin{pmatrix} 0 & -a & -b \\ a & 0 & -c \\ b & c & 0 \end{pmatrix}$$

Now,

$$(A+A') = \begin{pmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{pmatrix} + \begin{pmatrix} 0 & -a & -b \\ a & 0 & -c \\ b & c & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Therefore,

$$\frac{1}{2} (A+A') = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Now,

$$(A-A') = \begin{pmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{pmatrix} - \begin{pmatrix} 0 & -a & -b \\ a & 0 & -c \\ b & c & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 2a & 2b \\ -2a & 0 & 2c \\ -2b & -2c & 0 \end{pmatrix}$$

Thus,

$$\frac{1}{2} (A-A') = \begin{pmatrix} 0 & 2a & 2b \\ -2a & 0 & 2c \\ -2b & -2c & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{pmatrix}$$

Q

42

Expressing
matrices as
symmetric
plus skew-
symmetric
parts

(())

Express the following matrices as the sum of a symmetric and a skew symmetric matrix:

Part (i)

$$\begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}$$

Part (ii)

$$\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

Part (iii)

$$\begin{bmatrix} 3 & 3 & -1 \\ -2 & -2 & 1 \\ -4 & -5 & 2 \end{bmatrix}$$

Part (iv)

$$\begin{bmatrix} 1 & 5 \\ -1 & 2 \end{bmatrix}$$

Choose the correct answer in the Exercises 11 and 12 .

Given

(i)

Substitute

$$\text{Let } A = \begin{pmatrix} 3 & 5 \\ 1 & -1 \end{pmatrix}$$

Hence,

$$A' = \begin{pmatrix} 3 & 1 \\ 5 & -1 \end{pmatrix}$$

Now,

$$(A+A') = \begin{pmatrix} 3 & 5 \\ 1 & -1 \end{pmatrix} + \begin{pmatrix} 3 & 1 \\ 5 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 6 & 6 \\ 6 & -2 \end{pmatrix}$$

Let

$$P = \frac{1}{2} (A+A')$$

$$= \frac{1}{2} \begin{pmatrix} 6 & 6 \\ 6 & -2 \end{pmatrix}$$

$$= \begin{pmatrix} 3 & 3 \\ 3 & -1 \end{pmatrix}$$

Now,

$$P' = \begin{pmatrix} 3 & 3 \\ 3 & -1 \end{pmatrix}$$

$$= P$$

Thus, $P = \frac{1}{2} (A + A')$ is a symmetric matrix.

Now,

$$(A - A') = \begin{pmatrix} 3 & 5 \\ 1 & -1 \end{pmatrix} - \begin{pmatrix} 3 & 1 \\ 5 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 4 \\ -4 & 0 \end{pmatrix}$$

Let

$$Q = \frac{1}{2} (A - A')$$

$$= \frac{1}{2} \begin{pmatrix} 0 & 4 \\ -4 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$$

Now,

$$Q' = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$$

$$= -Q$$

Thus, $Q = \frac{1}{2} (A - A')$ is a skew symmetric matrix.

Conclusion

Representing A as the sum of P and Q :

$$P + Q = \begin{pmatrix} 3 & 3 \\ 3 & -1 \end{pmatrix} + \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 3 & 5 \\ 1 & -1 \end{pmatrix}$$

$$= A$$

Substitute

(ii)

Let

$$A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$$

Hence,

$$A' = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$$

Now,

$$(A + A') = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix} + \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix} = \begin{pmatrix} 12 & -4 & 4 \\ -4 & 6 & -2 \\ 4 & -2 & 6 \end{pmatrix}$$

Let

$$\begin{aligned}
 P &= \frac{1}{2} (A+A') \\
 &= \frac{1}{2} \begin{pmatrix} 12 & -4 & 4 \\ -4 & 6 & -2 \\ 4 & -2 & 6 \end{pmatrix} \\
 &= \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}
 \end{aligned}$$

Now,

$$\begin{aligned}
 P' &= \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix} \\
 &= P
 \end{aligned}$$

Thus, $P = \frac{1}{2} (A+A')$ is a symmetric matrix.

Now,

$$\begin{aligned}
 (A-A') &= \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix} - \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix} \\
 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}
 \end{aligned}$$

Let

$$\begin{aligned}
 Q &= \frac{1}{2} (A-A') \\
 &= \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}
 \end{aligned}$$

Now,

$$\begin{aligned}
 Q' &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
 &= -Q
 \end{aligned}$$

Thus, $Q = \frac{1}{2} (A-A')$ is a skew symmetric matrix.

Conclusion

Representing A as the sum of P and Q :

$$\begin{aligned}
 P+Q &= \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
 &= \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix} \\
 &= A
 \end{aligned}$$

Substitute

(iii)

Let

$$A = \begin{pmatrix} 3 & 3 & -1 \\ -2 & -2 & 1 \\ -4 & -5 & 2 \end{pmatrix}$$

Hence,

$$A' = \begin{pmatrix} 3 & -2 & -4 \\ 3 & -2 & -5 \\ -1 & 1 & 2 \end{pmatrix}$$

Now,

$$\begin{aligned} (A+A') &= \begin{pmatrix} 3 & 3 & -1 \\ -2 & -2 & 1 \\ -4 & -5 & 2 \end{pmatrix} + \begin{pmatrix} 3 & -2 & -4 \\ 3 & -2 & -5 \\ -1 & 1 & 2 \end{pmatrix} \\ &= \begin{pmatrix} 6 & 1 & -5 \\ 1 & -4 & -4 \\ -5 & -4 & 4 \end{pmatrix} \end{aligned}$$

Let

$$\begin{aligned} P &= \frac{1}{2} (A+A') \\ &= \frac{1}{2} \begin{pmatrix} 6 & 1 & -5 \\ 1 & -4 & -4 \\ -5 & -4 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 3 & \frac{1}{2} & \frac{-5}{2} \\ \frac{1}{2} & -2 & -2 \\ \frac{-5}{2} & -2 & 2 \end{pmatrix} \end{aligned}$$

Now,

$$\begin{aligned} P' &= \begin{pmatrix} 3 & \frac{1}{2} & \frac{-5}{2} \\ \frac{1}{2} & -2 & -2 \\ \frac{-5}{2} & -2 & 2 \end{pmatrix} \\ &= P \end{aligned}$$

Thus, $P = \frac{1}{2} (A+A')$ is a symmetric matrix.

Now,

$$\begin{aligned} (A-A') &= \begin{pmatrix} 3 & 3 & -1 \\ -2 & -2 & 1 \\ -4 & -5 & 2 \end{pmatrix} - \begin{pmatrix} 3 & -2 & -4 \\ 3 & -2 & -5 \\ -1 & 1 & 2 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 5 & 3 \\ -5 & 0 & 6 \\ -3 & -6 & 0 \end{pmatrix} \end{aligned}$$

Let

$$\begin{aligned}
 Q &= \frac{1}{2} (A-A') \\
 &= \frac{1}{2} \begin{pmatrix} 0 & 5 & 3 \\ -5 & 0 & 6 \\ -3 & -6 & 0 \end{pmatrix} \\
 &= \begin{pmatrix} 0 & \frac{5}{2} & \frac{3}{2} \\ \frac{-5}{2} & 0 & 3 \\ \frac{-3}{2} & -3 & 0 \end{pmatrix}
 \end{aligned}$$

Now,

$$\begin{aligned}
 Q' &= \begin{pmatrix} 0 & \frac{-5}{2} & \frac{-3}{2} \\ \frac{5}{2} & 0 & -3 \\ \frac{3}{2} & 3 & 0 \end{pmatrix} \\
 &= -Q
 \end{aligned}$$

Thus, $Q = \frac{1}{2} (A-A')$ is a skew symmetric matrix.

Conclusion

Representing A as the sum of P and Q :

$$\begin{aligned}
 P+Q &= \begin{pmatrix} 3 & \frac{1}{2} & \frac{-5}{2} \\ \frac{1}{2} & -2 & -2 \\ \frac{-5}{2} & -2 & 2 \end{pmatrix} + \begin{pmatrix} 0 & \frac{5}{2} & \frac{3}{2} \\ \frac{-5}{2} & 0 & 3 \\ \frac{-3}{2} & -3 & 0 \end{pmatrix} \\
 &= \begin{pmatrix} 3 & 3 & -1 \\ -2 & -2 & 1 \\ -4 & -5 & 2 \end{pmatrix} \\
 &= A
 \end{aligned}$$

Substitute

(iv)

$$\text{Let } A = \begin{pmatrix} 1 & 5 \\ -1 & 2 \end{pmatrix}$$

Hence,

$$A' = \begin{pmatrix} 1 & -1 \\ 5 & 2 \end{pmatrix}$$

Now,

$$\begin{aligned}
 (A+A') &= \begin{pmatrix} 1 & 5 \\ -1 & 2 \end{pmatrix} + \begin{pmatrix} 1 & -1 \\ 5 & 2 \end{pmatrix} \\
 &= \begin{pmatrix} 2 & 4 \\ 4 & 4 \end{pmatrix}
 \end{aligned}$$

Let

$$\begin{aligned}
 P &= \frac{1}{2} (A+A') \\
 &= \frac{1}{2} \begin{pmatrix} 2 & 4 \\ 4 & 4 \end{pmatrix} \\
 &= \begin{pmatrix} 1 & 2 \\ 2 & 2 \end{pmatrix}
 \end{aligned}$$

Now,

$$\begin{aligned}
 P' &= \begin{pmatrix} 1 & 2 \\ 2 & 2 \end{pmatrix} \\
 &= P
 \end{aligned}$$

Thus, $P = \frac{1}{2} (A+A')$ is a symmetric matrix.

Now,

$$\begin{aligned}
 (A-A') &= \begin{pmatrix} 1 & 5 \\ -1 & 2 \end{pmatrix} - \begin{pmatrix} 1 & -1 \\ 5 & 2 \end{pmatrix} \\
 &= \begin{pmatrix} 0 & 6 \\ -6 & 0 \end{pmatrix}
 \end{aligned}$$

Let

$$\begin{aligned}
 Q &= \frac{1}{2} (A-A') \\
 &= \frac{1}{2} \begin{pmatrix} 0 & 6 \\ -6 & 0 \end{pmatrix} \\
 &= \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix}
 \end{aligned}$$

Now,

$$\begin{aligned}
 Q' &= \begin{pmatrix} 0 & -3 \\ 3 & 0 \end{pmatrix} \\
 &= -Q
 \end{aligned}$$

Thus, $Q = \frac{1}{2} (A-A')$ is a skew symmetric matrix.

Conclusion

Representing A as the sum of P and Q :

$$\begin{aligned}
 P+Q &= \begin{pmatrix} 1 & 2 \\ 2 & 2 \end{pmatrix} + \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix} \\
 &= \begin{pmatrix} 1 & 5 \\ -1 & 2 \end{pmatrix} \\
 &= A
 \end{aligned}$$

Q

43

MCQ - nature
of AB-BA for
symmetric A,
B



If A, B are symmetric matrices of same order, then AB-BA is a (A) Skew symmetric matrix (B) Symmetric matrix (C) Zero matrix (D) Identity matrix

Given



If A and B are symmetric matrices of the same order, then

$$A' = A \text{ and } B' = B$$

Now consider,

$$\begin{aligned}
 (A B - B A)' &= (A B)' - (B A)' [\because (A-B)' = A' - B'] \\
 &= B' A' - A' B [\because (A B)' = B' A'] \\
 &= B A - A B \text{ [from (1)]} \\
 &= -(A B - B A)
 \end{aligned}$$

Therefore,

$$(A - B - A)' = -(A - B - A)$$

Conclusion

Thus, $A - B - A$ is a skew symmetric matrix.

The Correct option is A.

Q

44

MCQ - value of alpha given $A + A' = I$



If $A = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$, and $A + A' = I$, then the value of α is (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) π (D) $\frac{3\pi}{2}$

Given

It is given that $A = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$

Hence,

$$A' = \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix}$$

Now,

$$A + A' = I$$

Therefore,

$$\begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix} + \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 2 \cos \alpha & 0 \\ 0 & 2 \cos \alpha \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Comparing the corresponding elements of the two matrices, we have:

$$\Rightarrow 2 \cos \alpha = 1$$

$$\Rightarrow \cos \alpha = \frac{1}{2}$$

$$\Rightarrow \alpha = \cos^{-1} \frac{1}{2}$$

$$\Rightarrow \alpha = \frac{\pi}{3}$$

Conclusion

Thus, the correct option is B.

Q

45

MCQ - condition for A and B to be inverses



Matrices A and B will be inverse of each other only if (A) $AB = BA$ (B) $AB = BA = 0$ (C) $AB = 0, BA = I$ (D) $AB = BA = I$

Given

Let A be a square matrix of order m.

Suppose there exists another square matrix B of the same order m, such that $AB = BA = I$.

Then B is said to be the inverse of A.

In this case, it is clear that A is the inverse of B.

Thus, matrices A and B will be inverses of each other only if $AB = BA = I$.

Conclusion

The correct option is D.

Q**46**

Skew
symmetry of
 $AB-BA$ for
symmetric A ,
 B

(())

If A and B are symmetric matrices, prove that $AB-BA$ is a skew symmetric matrix.

Given

We are given that A and B are symmetric matrices.

So we have:

$$A' = A \text{ and } B' = B$$

Substitute

Now,

$$\begin{aligned}(A B-B A)' &= (A B)'-(B A)' [(A-B)' = A'-B'] \\&= B' A'-A' B' [(A B)' = B' A'] \\&= B A-A B \text{ [Using (1)]} \\&= -(A B-B A)\end{aligned}$$

Hence,

$$(A B-B A)' = -(A B-B A)$$

Conclusion

Thus, $A B-B A$ is a skew symmetric matrix.

Q**47**

Symmetry of
 $B'AB$

(*)

Show that the matrix $B'AB$ is symmetric or skew symmetric according as A is symmetric or skew symmetric.

Given

Suppose A is a symmetric matrix.

Then

$$A' = A$$

Consider,

$$\begin{aligned}(B' A B)' &= \{B' (A B)\}' \\&= (A B)' (B')' [\because (A B)' = B' A'] \\&= B' A' (B) [\because (B')' = B] \\&= B' (A' B) \\&= B' (A B) \text{ [Using (1)]}\end{aligned}$$

So,

$$(B' A B)' = B' A B$$

Thus, if A is a symmetric matrix, then $B' A B$ is a symmetric matrix.

Substitute

Now suppose A is a skew symmetric matrix.

Then

$$A' = -A$$

Consider,

$$\begin{aligned}(B' A B)' &= \{B' (A B)\}' \\&= (A B)' (B')' \\&= (B' A') B \\&= B' (-A) B \text{ [Using (2)]} \\&= -B' A B\end{aligned}$$

So,

$$(B' A B)' = -B' A B$$

Thus, if A is a skew symmetric matrix, then $B' A B$ is a skew symmetric matrix.

Conclusion

Hence, if A is symmetric or skew symmetric, then $B' A B$ is symmetric or skew symmetric accordingly.

Q

48

Solving for x,
y, z given
 $A'A=I$



Find the values of x, y, z if the matrix $A = \begin{bmatrix} 0 & 2y & z \\ x & y & -z \\ x & -y & z \end{bmatrix}$ satisfy the equation

$$A' A = I.$$

Given

We are given that

$$A = \begin{bmatrix} 0 & 2y & z \\ x & y & -z \\ x & -y & z \end{bmatrix}$$

So,

$$A' = \begin{bmatrix} 0 & x & x \\ 2y & y & -y \\ z & -z & z \end{bmatrix}$$

Substitute

$$\text{Now, } A' A = I$$

Hence,

$$\begin{aligned} \Rightarrow & \begin{bmatrix} 0 & x & x \\ 2y & y & -y \\ z & -z & z \end{bmatrix} \begin{bmatrix} 0 & 2y & z \\ x & y & -z \\ x & -y & z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ \Rightarrow & \begin{bmatrix} 0+x^2+x^2 & 0+xy-xy & 0-xz+xz \\ 0+xy-xy & 4y^2+y^2+y^2 & 2yz-yz-yz \\ 0-xz+xz & 2yz-yz-yz & z^2+z^2+z^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ \Rightarrow & \begin{bmatrix} 2x^2 & 0 & 0 \\ 0 & 6y^2 & 0 \\ 0 & 0 & 3z^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

Comparing the corresponding elements, we get:

$$2x^2 = 1$$

$$\Rightarrow x = \pm \frac{1}{\sqrt{2}}$$

$$6y^2 = 1$$

$$\Rightarrow y = \pm \frac{1}{\sqrt{6}}$$

$$3z^2 = 1$$

$$\Rightarrow z = \pm \frac{1}{\sqrt{3}}$$

Conclusion

$$\text{Thus, } x = \pm \frac{1}{\sqrt{2}}, y = \pm \frac{1}{\sqrt{6}} \text{ and } z = \pm \frac{1}{\sqrt{3}}$$

Q

49

Solving a
matrix
product
equation for
x



For what values of x : $\begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ x \end{bmatrix} = \mathbf{0} ?$

Key formula

We have:

$$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ x \end{bmatrix} = 0$$

Substitute

So,

$$\Rightarrow \begin{bmatrix} 1+4+1 & 2+0+0 & 0+2+2 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ x \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 6 & 2 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ x \end{bmatrix} = 0$$

$$\Rightarrow [6(0)+2(2)+4(x)] = 0$$

$$\Rightarrow [4+4x] = 0$$

$$\Rightarrow 4x = -4$$

$$\Rightarrow x = -1$$

Conclusion

Thus, the required value of x = -1.

Q

50

Verifying a
matrix
polynomial
identity

(())

If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, show that $A^2 - 5A + 7I = 0$.

Given

We are given that $A = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}$

Substitute

So,

$$\begin{aligned} A^2 &= A \cdot A \\ &= \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix} \\ &= \begin{pmatrix} 3(3)+1(-1) & 3(1)+1(2) \\ -1(3)+2(-1) & -1(1)+2(2) \end{pmatrix} \\ &= \begin{pmatrix} 8 & 5 \\ -5 & 3 \end{pmatrix} \end{aligned}$$

Now,

$$\begin{aligned} \text{L H S} &= A^2 - 5A + 7I \\ &= \begin{pmatrix} 8 & 5 \\ -5 & 3 \end{pmatrix} - 5 \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix} + 7 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 8 & 5 \\ -5 & 3 \end{pmatrix} - \begin{pmatrix} 15 & 5 \\ -5 & 10 \end{pmatrix} + \begin{pmatrix} 7 & 0 \\ 0 & 7 \end{pmatrix} \\ &= \begin{pmatrix} -7 & 0 \\ 0 & -7 \end{pmatrix} + \begin{pmatrix} 7 & 0 \\ 0 & 7 \end{pmatrix} \\ &= 0 \\ &= \text{RHS} \end{aligned}$$

Conclusion

Thus, $A^2 - 5A + 7I = 0$

Q

51

Solving a
matrix
product
equation for
x



Find x, if $\begin{bmatrix} x & -5 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ 4 \\ 1 \end{bmatrix} = \mathbf{0}$

Key formula

We have

$$\begin{bmatrix} x & -5 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ 4 \\ 1 \end{bmatrix} = 0$$

Substitute

So,

$$\Rightarrow \begin{bmatrix} x+0-2 & 0-10+0 & 2x-5-3 \end{bmatrix} \begin{bmatrix} x \\ 4 \\ 1 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} x-2 & -10 & 2x-8 \end{bmatrix} \begin{bmatrix} x \\ 4 \\ 1 \end{bmatrix} = 0$$

$$\Rightarrow [x(x-2)-40+2x-8] = 0$$

$$\Rightarrow [x^2-2x-40+2x-8] = 0$$

$$\Rightarrow [x^2-48] = 0$$

$$\Rightarrow x^2-48 = 0$$

$$\Rightarrow x^2 = 48$$

$$\Rightarrow x = \pm 4\sqrt{3}$$

Conclusion

Thus, $x = \pm 4\sqrt{3}$.

Q

52

Matrix
algebra -
sales
revenue and
profit



A manufacturer produces three products x, y, z which he sells in two markets. Annual sales are indicated below:

Market	Products		
I	10,000	2,000	18,000
II	6,000	20,000	8,000

Part (a)

If unit sale prices of x, y and z are ₹ 2.50 , ₹ 1.50 and ₹ 1.00 , respectively, find the total revenue in each market with the help of matrix algebra.

Part (b)

If the unit costs of the above three commodities are ₹ 2.00 , ₹ 1.00 and 50 paise respectively. Find the gross profit.

Given

(a) The unit sale prices of x, y and z are ₹ 2.50 , ₹ 1.50 and ₹ 1.00 respectively.

So, the total revenue in market I can be written as a matrix:

$$\begin{bmatrix} 10000 & 2000 & 18000 \end{bmatrix} \begin{bmatrix} 2.50 \\ 1.50 \\ 1.00 \end{bmatrix} = 10000 \times 2.50 + 2000 \times 1.50 + 18000 \times 1.00$$

$$= 25000 + 3000 + 18000$$

$$= 46000$$

The total revenue in market II can be written as a matrix:

$$\begin{bmatrix} 6000 & 20000 & 8000 \end{bmatrix} \begin{bmatrix} 2.50 \\ 1.50 \\ 1.00 \end{bmatrix} = 6000 \times 2.50 + 20000 \times 1.50 + 8000 \times 1.00$$

$$= 15000 + 30000 + 8000$$

$$= 53000$$

Thus, the total revenue in market I is ₹46000 and the total revenue in market II is ₹53000.

Conclusion

(b) The unit costs of x, y and z are ₹ 2.00 , ₹ 1.00 and 50 paise respectively.

So, the total cost of all the products in market I can be written as a matrix:

$$\begin{bmatrix} 10000 & 2000 & 18000 \end{bmatrix} \begin{bmatrix} 2.00 \\ 1.00 \\ 0.50 \end{bmatrix} = 10000 \times 2.00 + 2000 \times 1.00 + 18000 \times 0.50 \\ = 20000 + 2000 + 9000 \\ = 31000$$

Since the total revenue in market I is ₹ 46000 , the gross profit in this market in ₹ is $46000 - 31000 = 15000$

The total cost of all the products in market II can be written as a matrix:

$$\begin{bmatrix} 6000 & 20000 & 8000 \end{bmatrix} \begin{bmatrix} 2.00 \\ 1.00 \\ 0.50 \end{bmatrix} = 6000 \times 2.00 + 20000 \times 1.00 + 8000 \times 0.50 \\ = 12000 + 20000 + 4000 \\ = 36000$$

Since the total revenue in market I is ₹ 53000, the gross profit in this market in ₹ is $53000 - 36000 = 17000$

Thus, the gross profit in market I is ₹15000 and in market II is ₹17000.

Q

53

Finding a matrix X from a matrix equation



Find the matrix X so that $X \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} -7 & -8 & -9 \\ 2 & 4 & 6 \end{bmatrix}$ Choose the correct answer in the following questions:

Given

We are given that $X \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} -7 & -8 & -9 \\ 2 & 4 & 6 \end{bmatrix}$

The matrix on the R.

H.

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of the equation is a 2×3 matrix.

The one on the L.

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of the equation is a 2×3 matrix.

So, X has to be a 2×2 matrix.

Key formula

Now, let

$$X = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

So,

$$\Rightarrow \begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} -7 & -8 & -9 \\ 2 & 4 & 6 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} a+4c & 2a+5c & 3a+6c \\ b+4d & 2b+5d & 3b+6d \end{bmatrix} = \begin{bmatrix} -7 & -8 & -9 \\ 2 & 4 & 6 \end{bmatrix}$$

Substitute

Equating the corresponding elements of the two matrices, we get:

$$a+4c = -7 \quad 2a+5c = -8 \quad 3a+6c = -9$$

$$b+4d = 2 \quad 2b+5d = 4 \quad 3b+6d = 6$$

Now,

$$a+4c = -7$$

$$\Rightarrow a = -7-4c$$

So,

$$2a+5c = -8$$

$$\Rightarrow 2(-7-4c)+5c = -8$$

$$\Rightarrow -14-8c+5c = -8$$

$$\Rightarrow -3c = 6$$

$$\Rightarrow c = -2$$

Hence,

$$\Rightarrow a = -7-4(-2)$$

$$\Rightarrow a = -7+8$$

$$\Rightarrow a = 1$$

Now,

$$b+4d = 2$$

$$\Rightarrow b = 2-4d$$

So,

$$2b+5d = 4$$

$$\Rightarrow 2(2-4d)+5d = 4$$

$$\Rightarrow 4-8d+5d = 4$$

$$\Rightarrow -3d = 0$$

$$\Rightarrow d = 0$$

Hence,

$$b = 2-4d$$

$$\Rightarrow b = 2$$

Conclusion

Thus, $a = 1$, $b = 2$, $c = -2$ and $d = 0$

Hence, the required matrix $X = \begin{bmatrix} 1 & -2 \\ 2 & 0 \end{bmatrix}$

Q

54

MCQ -
condition on
alpha, beta,
gamma given
 $A^2=I$

(())

If $A = \begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix}$ is such that $A^2 = I$, then (A) $1+\alpha^2+\beta\gamma = 0$ (B) $1-\alpha^2+\beta\gamma = 0$ (C) $1-\alpha^2-\beta\gamma = 0$ (D) $1+\alpha^2-\beta\gamma = 0$

Given

We are given that $A = \begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix}$

Substitute

So,

$$\begin{aligned}
 A^2 &= A \cdot A \\
 &= \begin{pmatrix} \alpha & \beta \\ \gamma & -\alpha \end{pmatrix} \begin{pmatrix} \alpha & \beta \\ \gamma & -\alpha \end{pmatrix} \\
 &= \begin{pmatrix} \alpha^2 + \beta\gamma & \alpha\beta - \alpha\beta \\ \alpha\gamma - \alpha\gamma & \beta\gamma + \alpha^2 \end{pmatrix} \\
 &= \begin{pmatrix} \alpha^2 + \beta\gamma & 0 \\ 0 & \beta\gamma + \alpha^2 \end{pmatrix}
 \end{aligned}$$

Now, $A^2 = I$

Hence,

$$\begin{pmatrix} \alpha^2 + \beta\gamma & 0 \\ 0 & \beta\gamma + \alpha^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Comparing the corresponding elements, we get:

$$\begin{aligned}
 \alpha^2 + \beta\gamma &= 1 \\
 \Rightarrow \alpha^2 + \beta\gamma - 1 &= 0 \\
 \Rightarrow 1 - \alpha^2 - \beta\gamma &= 0
 \end{aligned}$$

Conclusion

Thus, the correct option is C.

Q

55

MCQ - matrix
both
symmetric
and skew
symmetric



If the matrix A is both symmetric and skew symmetric, then (A) A is a diagonal matrix (B) A is a zero matrix (C) A is a square matrix (D) None of these

Given

If the matrix A is both symmetric and skew symmetric, then

$$A' = A \text{ and } A' = -A$$

Substitute

Hence,

$$\begin{aligned}
 \Rightarrow A &= -A \\
 \Rightarrow A + A &= 0 \\
 \Rightarrow 2A &= 0 \\
 \Rightarrow A &= 0
 \end{aligned}$$

So, A is a zero matrix.

Conclusion

Thus, the correct option is B.

Q

56

MCQ -
simplifying
(I+A)³-7A
given A²=A



If A is square matrix such that $A^2 = A$, then $(I+A)^3 - 7A$ is equal to (A) A (B) I-A (C) I (D) 3A

Given

We are given that A is a square matrix such that $A^2 = A$.

Substitute

Now,

$$\begin{aligned}
(I+A)^3 - 7A &= I^3 + A^3 + 3I^2A + 3IA^2 + 3A^2I - 7A \\
&= I + A^2 \cdot A + 3A + 3A^2 - 7A \\
&= I + A \cdot A + 3A + 3A - 7A \quad [\because A^2 = A] \\
&= I + A^2 - A \\
&= I + A - A \quad [\because A^2 = A] \\
&= I
\end{aligned}$$

Hence,

$$(I+A)^3 - 7A = I$$

Conclusion

Thus, the correct option is C.

Q

OLD-1

Inverse via elementary row operations



Using elementary transformation, Find the inverse of the matrix $\begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}$, if exists.

Conclusion

$$\text{Let } A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}$$

We know that $A = I A$

Therefore,

$$\begin{aligned} \Rightarrow \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} A \\ \Rightarrow \begin{pmatrix} 1 & -1 \\ 0 & 5 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} A \quad (R_2 \rightarrow R_2 - 2R_1) \\ \Rightarrow \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} A \quad (R_2 \rightarrow \frac{1}{5} R_2) \\ \Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} &= \begin{pmatrix} 3 & 1 \\ 5 & 5 \end{pmatrix} A \quad (R_1 \rightarrow R_1 + R_2) \\ \Rightarrow A^{-1} &= \begin{pmatrix} 3 & 1 \\ 5 & 5 \end{pmatrix} \end{aligned}$$

Q

OLD-2

Inverse via elementary row operations



Using elementary transformation, Find the inverse of the matrix $\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, if exists.

Conclusion

$$\text{Let } A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$$

We know that $A = I A$

Therefore,

$$\begin{aligned} \Rightarrow \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} A \\ \Rightarrow \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} &= \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} A \quad (R_1 \rightarrow R_1 - R_2) \\ \Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} &= \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} A \\ \Rightarrow A^{-1} &= \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \end{aligned}$$

Q

OLD-
3Inverse via
elementary
row
operations

Using elementary transformation, Find the inverse of the matrix $\begin{pmatrix} 1 & 3 \\ 2 & 7 \end{pmatrix}$, if exists.

Conclusion

$$\text{Let } A = \begin{pmatrix} 1 & 3 \\ 2 & 7 \end{pmatrix}$$

We know that $A = I A$

Therefore,

$$\begin{aligned} \Rightarrow \begin{pmatrix} 1 & 3 \\ 2 & 7 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} A \\ \Rightarrow \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} A \quad (R_2 \rightarrow R_2 - 2R_1) \\ \Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} &= \begin{pmatrix} 7 & -3 \\ -2 & 1 \end{pmatrix} A \quad (R_1 \rightarrow R_1 - 3R_2) \\ \Rightarrow A^{-1} &= \begin{pmatrix} 7 & -3 \\ -2 & 1 \end{pmatrix} \end{aligned}$$

Q

OLD-
4Inverse via
elementary
row
operations

Using elementary transformation, Find the inverse of the matrix $\begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix}$, if exists.

Conclusion

$$\text{Let } A = \begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix}$$

We know that $A = I A$

Therefore,

$$\begin{aligned} \Rightarrow \begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} A \\ \Rightarrow \begin{pmatrix} 1 & \frac{3}{2} \\ 5 & 7 \end{pmatrix} &= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix} A \\ \Rightarrow \begin{pmatrix} 1 & \frac{3}{2} \\ 0 & \frac{-1}{2} \end{pmatrix} &= \begin{pmatrix} \frac{1}{2} & 0 \\ \frac{-5}{2} & 1 \end{pmatrix} A \\ \Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & \frac{-1}{2} \end{pmatrix} &= \begin{pmatrix} -7 & 3 \\ \frac{-5}{2} & 1 \end{pmatrix} A \\ \Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} &= \begin{pmatrix} -7 & 3 \\ 5 & -2 \end{pmatrix} A \\ \Rightarrow A^{-1} &= \begin{pmatrix} -7 & 3 \\ 5 & -2 \end{pmatrix} \end{aligned}$$

Q**OLD-5**

Inverse via elementary row operations



Using elementary transformation, Find the inverse of the matrix $\begin{pmatrix} 2 & 1 \\ 7 & 4 \end{pmatrix}$, if exists.

Conclusion

$$\text{Let } A = \begin{pmatrix} 2 & 1 \\ 7 & 4 \end{pmatrix}$$

We know that $A = I A$

Therefore,

$$\begin{aligned} \Rightarrow \begin{pmatrix} 2 & 1 \\ 7 & 4 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} A \\ \Rightarrow \begin{pmatrix} 1 & \frac{1}{2} \\ 7 & 4 \end{pmatrix} &= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix} A \\ \Rightarrow \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & \frac{1}{2} \end{pmatrix} &= \begin{pmatrix} \frac{1}{2} & 0 \\ \frac{-7}{2} & 1 \end{pmatrix} A & \quad (R_2 \rightarrow R_2 - 7R_1) \cdot (R_2 \rightarrow 2R_1) \\ \Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix} &= \begin{pmatrix} 4 & -1 \\ \frac{-7}{2} & 1 \end{pmatrix} A \\ \Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} &= \begin{pmatrix} 4 & -1 \\ -7 & 2 \end{pmatrix} A \\ \Rightarrow A^{-1} &= \begin{pmatrix} 4 & -1 \\ -7 & 2 \end{pmatrix} \end{aligned}$$

Q**OLD-6**

Inverse via elementary row operations

(())

Using elementary transformation, Find the inverse of the matrix $\begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}$, if exists.

Conclusion

$$\text{Let } A = \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}$$

We know that $A = I A$

Therefore,

$$\begin{aligned} \Rightarrow \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} A \\ \Rightarrow \begin{pmatrix} 1 & \frac{5}{2} \\ 1 & 3 \end{pmatrix} &= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix} A \\ \Rightarrow \begin{pmatrix} 1 & \frac{5}{2} \\ 0 & \frac{1}{2} \end{pmatrix} &= \begin{pmatrix} \frac{1}{2} & 0 \\ \frac{-1}{2} & 1 \end{pmatrix} A & \quad (R_2 \rightarrow R_2 - R_1) \cdot (R_1 \rightarrow 2R_1) \cdot (R_2 \rightarrow R_2) \\ \Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix} &= \begin{pmatrix} 3 & -5 \\ \frac{-1}{2} & 1 \end{pmatrix} A \\ \Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} &= \begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix} A \\ \Rightarrow A^{-1} &= \begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix} \end{aligned}$$

Q

OLD-
7Inverse via
elementary
row
operations

Using elementary transformation, Find the inverse of the matrix $\begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$, if exists.

Conclusion

$$\text{Let } A = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$$

We know that $A = I A$

Therefore,

$$\begin{aligned} \Rightarrow \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix} &= A \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ \Rightarrow \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} &= A \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} \quad (C_1 \rightarrow C_1 - 2 C_2) \\ \Rightarrow \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} &= A \begin{pmatrix} 1 & -1 \\ -2 & 3 \end{pmatrix} \quad (C_2 \rightarrow C_2 - C_1) \\ \Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} &= A \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix} \quad (C_1 \rightarrow C_1 - C_2) \\ \Rightarrow A^{-1} &= \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix} \end{aligned}$$

Q

OLD-
8Inverse via
elementary
row
operations

Using elementary transformation, Find the inverse of the matrix $\begin{pmatrix} 4 & 5 \\ 3 & 4 \end{pmatrix}$, if exists.

Conclusion

$$\text{Let } A = \begin{pmatrix} 4 & 5 \\ 3 & 4 \end{pmatrix}$$

We know that $A = I A$

Therefore,

$$\begin{aligned} \Rightarrow \begin{pmatrix} 4 & 5 \\ 3 & 4 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} A \\ \Rightarrow \begin{pmatrix} 1 & 1 \\ 3 & 4 \end{pmatrix} &= \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} A \quad (R_1 \rightarrow R_1 - R_2) \\ \Rightarrow \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} &= \begin{pmatrix} 1 & -1 \\ -3 & 4 \end{pmatrix} A \quad (R_2 \rightarrow R_2 - 3 R_1) \\ \Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} &= \begin{pmatrix} 4 & -5 \\ -3 & 4 \end{pmatrix} A \quad (R_1 \rightarrow R_1 - R_2) \\ \Rightarrow A^{-1} &= \begin{pmatrix} 4 & -5 \\ -3 & 4 \end{pmatrix} \end{aligned}$$

Q

OLD-
9Inverse via
elementary
row
operations

Using elementary transformation, Find the inverse of the matrix $\begin{pmatrix} 3 & 10 \\ 2 & 7 \end{pmatrix}$, if exists.

Conclusion

$$\text{Let } A = \begin{pmatrix} 3 & 10 \\ 2 & 7 \end{pmatrix}$$

We know that $A = I A$

Therefore,

$$\Rightarrow \begin{pmatrix} 3 & 10 \\ 2 & 7 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} A$$

$$\Rightarrow \begin{pmatrix} 1 & 3 \\ 2 & 7 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} A \quad (R_1 \rightarrow R_1 - R_2)$$

$$\Rightarrow \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -2 & 3 \end{pmatrix} A \quad (R_2 \rightarrow R_2 - 2 R_1)$$

$$\Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 7 & -10 \\ -2 & 3 \end{pmatrix} A \quad (R_1 \rightarrow R_1 - 3 R_2)$$

$$\Rightarrow A^{-1} = \begin{pmatrix} 7 & -10 \\ -2 & 3 \end{pmatrix}$$

Q

OLD-10

Inverse via elementary row operations

(())

Using elementary transformation, Find the inverse of the matrix $\begin{pmatrix} 3 & -1 \\ -4 & 2 \end{pmatrix}$, if exists.

Conclusion

$$\text{Let } A = \begin{pmatrix} 3 & -1 \\ -4 & 2 \end{pmatrix}$$

We know that $A = I A$

Therefore,

$$\Rightarrow \begin{pmatrix} 3 & -1 \\ -4 & 2 \end{pmatrix} = A \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix} = A \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \quad (C_1 \rightarrow C_1 + 2 C_2)$$

$$\Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} = A \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} \quad (C_2 \rightarrow C_2 + C_1)$$

$$\Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = A \begin{pmatrix} 1 & \frac{1}{2} \\ 2 & \frac{3}{2} \end{pmatrix} \quad (C_2 \rightarrow \frac{1}{2} C_2)$$

$$\Rightarrow A^{-1} = \begin{pmatrix} 1 & \frac{1}{2} \\ 2 & \frac{3}{2} \end{pmatrix}$$

Q

OLD-11

Inverse via elementary row operations

(())

Using elementary transformation, Find the inverse of the matrix $\begin{pmatrix} 2 & -6 \\ 1 & -2 \end{pmatrix}$, if exists.

Conclusion

$$\text{Let } A = \begin{pmatrix} 2 & -6 \\ 1 & -2 \end{pmatrix}$$

We know that $A = I A$

Therefore,

$$\begin{aligned}
&\Rightarrow \begin{pmatrix} 2 & -6 \\ 1 & -2 \end{pmatrix} = A \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\
&\Rightarrow \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} = A \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} \quad (C_2 \rightarrow C_2 + 3 C_1) \\
&\Rightarrow \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = A \begin{pmatrix} -2 & 3 \\ -1 & 1 \end{pmatrix} \quad (C_1 \rightarrow C_1 - C_2) \\
&\Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = A \begin{pmatrix} -1 & 3 \\ -\frac{1}{2} & 1 \end{pmatrix} \quad (C_1 \rightarrow \frac{1}{2} C_1) \\
&\Rightarrow A^{-1} = \begin{pmatrix} -1 & 3 \\ -\frac{1}{2} & 1 \end{pmatrix}
\end{aligned}$$

Q

OLD-12

Inverse via elementary row operations



Using elementary transformation, Find the inverse of the matrix $\begin{pmatrix} 6 & -3 \\ -2 & 1 \end{pmatrix}$, if exists.

Given

Let

$$A = \begin{pmatrix} 6 & -3 \\ -2 & 1 \end{pmatrix}$$

We know that $A = I A$

Therefore,

$$\begin{aligned}
&\Rightarrow \begin{pmatrix} 6 & -3 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} A \\
&\Rightarrow \begin{pmatrix} 1 & -\frac{1}{2} \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{6} & 0 \\ 0 & 1 \end{pmatrix} A \quad (R_1 \rightarrow \frac{1}{6} R_1) \\
&\Rightarrow \begin{pmatrix} 1 & -\frac{1}{2} \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{6} & 0 \\ \frac{1}{3} & 1 \end{pmatrix} A \quad (R_2 \rightarrow R_2 + 2 R_1)
\end{aligned}$$

Conclusion

In this equation, all the entries in the second row of the matrix on the L.

H.

S.

are zero.

Thus, A^{-1} does not exist.

Q

OLD-13

Inverse via elementary row operations



Using elementary transformation, Find the inverse of the matrix $\begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}$, if exists.

Conclusion

$$\text{Let } A = \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}$$

We know that $A = I A$

Therefore,

$$\Rightarrow \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} A$$

$$\Rightarrow \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} A \quad (R_1 \rightarrow R_1 + R_2)$$

$$\Rightarrow \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} A \quad (R_2 \rightarrow R_2 + R_1)$$

$$\Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} A \quad (R_1 \rightarrow R_1 + R_2)$$

$$\Rightarrow A^{-1} = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$$

Q

OLD-14

Inverse via elementary row operations

(())

Using elementary transformation, Find the inverse of the matrix $\begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix}$, if exists.

Given

$$\text{Let } A = \begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix}$$

We know that $A = I A$

Therefore,

$$\Rightarrow \begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} A$$

$$\Rightarrow \begin{pmatrix} 0 & 0 \\ 4 & 2 \end{pmatrix} = \begin{pmatrix} 1 & \frac{-1}{2} \\ 0 & 1 \end{pmatrix} A \quad (R_1 \rightarrow R_1 - \frac{1}{2} R_2)$$

Conclusion

In this equation, all the entries in the first row of the matrix on the L.

H.

S.

are zero.

Thus, A^{-1} does not exist.

Q

OLD-15

Inverse via elementary row operations



Using elementary transformation, Find the inverse of the matrix $\begin{pmatrix} 2 & -3 & 3 \\ 2 & 2 & 3 \\ 3 & -2 & 2 \end{pmatrix}$, if exists.

Conclusion

Let

$$A = \begin{pmatrix} 2 & -3 & 3 \\ 2 & 2 & 3 \\ 3 & -2 & 2 \end{pmatrix}$$

We know that $A = I A$

Therefore,

$$\begin{aligned}
\Rightarrow \begin{pmatrix} 2 & -3 & 3 \\ 2 & 2 & 3 \\ 3 & -2 & 2 \end{pmatrix} &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} A \\
\Rightarrow \begin{pmatrix} 2 & -3 & 3 \\ 0 & 5 & 0 \\ 3 & -2 & 2 \end{pmatrix} &= \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} A (R_2 \rightarrow R_2 - R_1) \\
\Rightarrow \begin{pmatrix} 2 & -3 & 3 \\ 0 & 1 & 0 \\ 3 & -2 & 2 \end{pmatrix} &= \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} A (R_2 \rightarrow \frac{1}{5} R_2) \\
\Rightarrow \begin{pmatrix} -1 & -1 & 1 \\ 0 & 1 & 0 \\ 3 & -2 & 2 \end{pmatrix} &= \begin{pmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} A (R_1 \rightarrow R_1 - R_3) \\
\Rightarrow \begin{pmatrix} -1 & 0 & 1 \\ 0 & 1 & 0 \\ 3 & 0 & 2 \end{pmatrix} &= \begin{pmatrix} 4 & 1 & -1 \\ 5 & 5 & 0 \\ -1 & 1 & 0 \\ 5 & 5 & 1 \\ -2 & 2 & 1 \end{pmatrix} A (R_1 \rightarrow R_1 + R_2 \text{ and } R_3 \rightarrow R_3 + 2 R_2) \\
\Rightarrow \begin{pmatrix} -1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{pmatrix} &= \begin{pmatrix} 4 & 1 & -1 \\ 5 & 5 & 0 \\ -1 & 1 & 0 \\ 5 & 5 & -2 \end{pmatrix} A (R_3 \rightarrow R_3 + 3 R_1) \\
\Rightarrow \begin{pmatrix} -1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} &= \begin{pmatrix} 4 & 1 & -1 \\ 5 & 5 & 0 \\ -1 & 1 & 0 \\ 2 & 1 & -2 \end{pmatrix} A (R_3 \rightarrow \frac{1}{5} R_3) \\
\Rightarrow \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} &= \begin{pmatrix} 2 & 0 & -3 \\ 5 & 5 & 5 \\ -1 & 1 & 0 \\ 2 & 1 & -2 \end{pmatrix} A (R_1 \rightarrow R_1 - R_3) \\
\Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} &= \begin{pmatrix} -2 & 0 & 3 \\ 5 & 5 & 5 \\ -1 & 1 & 0 \\ 2 & 1 & -2 \end{pmatrix} A (R_1 \rightarrow (-1) R_1) \\
\Rightarrow A^{-1} &= \begin{pmatrix} \frac{-2}{5} & 0 & \frac{3}{5} \\ \frac{-1}{5} & \frac{1}{5} & 0 \\ \frac{2}{5} & \frac{1}{5} & \frac{-2}{5} \end{pmatrix}
\end{aligned}$$

Q

OLD-16

Inverse via elementary row operations



Using elementary transformation, Find the inverse of the matrix $\begin{pmatrix} 1 & 3 & -2 \\ -3 & 0 & -5 \\ 2 & 5 & 0 \end{pmatrix}$, if exists.

Conclusion

Let

$$A = \begin{pmatrix} 1 & 3 & -2 \\ -3 & 0 & -5 \\ 2 & 5 & 0 \end{pmatrix}$$

We know that $A = I A$

Therefore,

$$\begin{aligned} \Rightarrow \begin{pmatrix} 1 & 3 & -2 \\ -3 & 0 & -5 \\ 2 & 5 & 0 \end{pmatrix} &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} A \\ \Rightarrow \begin{pmatrix} 1 & 3 & -2 \\ 0 & 9 & -11 \\ 0 & -1 & 4 \end{pmatrix} &= \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix} A \\ \Rightarrow \begin{pmatrix} 1 & 0 & 10 \\ 0 & 1 & 21 \\ 0 & -1 & 4 \end{pmatrix} &= \begin{pmatrix} -5 & 0 & 3 \\ -13 & 1 & 8 \\ -2 & 0 & 1 \end{pmatrix} A \\ \Rightarrow \begin{pmatrix} 1 & 0 & 10 \\ 0 & 1 & 21 \\ 0 & 0 & 25 \end{pmatrix} &= \begin{pmatrix} -5 & 0 & 3 \\ -13 & 1 & 8 \\ -15 & 1 & 9 \end{pmatrix} A \\ \Rightarrow \begin{pmatrix} 1 & 0 & 10 \\ 0 & 1 & 21 \\ 0 & 0 & 1 \end{pmatrix} &= \left(\begin{array}{ccc|ccc} -5 & 0 & 3 & & & \\ -13 & 1 & 8 & & & \\ -3 & 1 & 9 & & & \\ \hline 5 & 25 & 25 & & & \end{array} \right) A \quad (R_2 + 3R_1 \text{ and } R_3 \rightarrow R_3 - 2R_1) \\ \Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} &= \left(\begin{array}{ccc|ccc} 1 & -2 & -3 & & & \\ & 5 & 5 & & & \\ -2 & 4 & 11 & & & \\ \hline 5 & 25 & 25 & & & \\ -3 & 1 & 9 & & & \\ \hline 5 & 25 & 25 & & & \end{array} \right) A \\ \Rightarrow A^{-1} &= (R_1 \rightarrow R_1 + 3R_3 \text{ and } R_2 \rightarrow R_2 + 8R_3) \\ &\quad \begin{array}{ccc|ccc} & -2 & 11 & & & \\ -2 & 5 & 25 & 1 & 9 & \\ \hline 5 & -3 & -3 & 25 & 25 & \\ & 25 & 5 & & 25 & \end{array} \end{aligned}$$

Q

OLD-17

Inverse via elementary row operations



Using elementary transformation, Find the inverse of the matrix $\begin{pmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{pmatrix}$, if exists.

Conclusion

Let

$$A = \begin{pmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{pmatrix}$$

We know that $A = I A$

Therefore,

$$\begin{aligned}
&\Rightarrow \begin{pmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} A \\
&\Rightarrow \begin{pmatrix} 1 & 0 & \frac{-1}{2} \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} A \\
&\Rightarrow \begin{pmatrix} 1 & 0 & \frac{-1}{2} \\ 0 & 1 & \frac{5}{2} \\ 0 & 1 & 3 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ \frac{-5}{2} & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} A \\
&\Rightarrow \begin{pmatrix} 1 & 0 & \frac{-1}{2} \\ 0 & 1 & \frac{5}{2} \\ 0 & 0 & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ \frac{-5}{2} & 1 & 0 \\ \frac{5}{2} & -1 & 1 \end{pmatrix} A \quad (R_3 \rightarrow R_3 - R_2) \quad (R_1 \rightarrow R_1 + R_3) \\
&\Rightarrow \begin{pmatrix} 1 & 0 & \frac{-1}{2} \\ 0 & 1 & \frac{5}{2} \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ \frac{-5}{2} & 1 & 0 \\ 5 & -2 & 2 \end{pmatrix} A \\
&\Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{pmatrix} A \\
&\frac{1}{2} R_3 \text{ and } R_2 \rightarrow R_2 - \frac{5}{2} R_3 \cdot (R_1) A^{-1} = \begin{pmatrix} 3 & 1 \\ -15 & -5 \\ 5 & -2 \end{pmatrix} \cdot (R_1) A
\end{aligned}$$

Q

**MISC
-1**

**Matrix
power by
induction**

(())

Let $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, show that $(aI + bA)^n = a^n I + n a^{n-1} b A$, where I is the identity matrix of order 2 and $n \in \mathbb{N}$.

Given

We are given that $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

Substitute

We will prove the result using the principle of mathematical induction.

For $n = 1$, we have:

$$P(1): (aI + bA) = aI + b a^0 A = aI + bA$$

So the result is true for $n = 1$.

Assume the result is true for $n = k$

$$\text{That is, } P(k): (aI + bA)^k = a^k I + k a^{k-1} b A$$

Conclusion

Now we prove that the result is true for $n = k+1$.

Consider,

$$\begin{aligned}
 (aI+bA)^{k+1} &= (aI+bA)^k(aI+bA) \\
 &= (a^k I + k a^{k-1} b A)(aI+bA) \\
 &= a^{k+1} I + k a^k b A + a^k b I + k a^{k-1} b^2 A^2 \\
 &= a^{k+1} I + (k+1) a^k b A + k a^{k-1} b^2 A^2
 \end{aligned}$$

Now,

$$A^2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = 0$$

From (1), we get

$$\begin{aligned}
 (aI+bA)^{k+1} &= a^{k+1} I + (k+1) a^k b A + 0 \\
 &= a^{k+1} I + (k+1) a^k b A
 \end{aligned}$$

So the result is true for $n = k+1$.

Thus, by the principle of mathematical induction, we have:

$$(aI+bA)^n = a^n I + n a^{n-1} b A \text{ where } A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, n \in \mathbb{N}$$

Q

**MISC
-2**

**Matrix
power by
induction**



If

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \text{ prove that } A^n = \begin{pmatrix} 3^{n-1} & 3^{n-1} & 3^{n-1} \\ 3^{n-1} & 3^{n-1} & 3^{n-1} \\ 3^{n-1} & 3^{n-1} & 3^{n-1} \end{pmatrix}, n \in \mathbb{N}$$

Given



We are given that

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

Substitute

We will prove the result using the principle of mathematical induction.

For $n = 1$, we have:

$$P(1): \begin{pmatrix} 3^{1-1} & 3^{1-1} & 3^{1-1} \\ 3^{1-1} & 3^{1-1} & 3^{1-1} \\ 3^{1-1} & 3^{1-1} & 3^{1-1} \end{pmatrix} = \begin{pmatrix} 3^0 & 3^0 & 3^0 \\ 3^0 & 3^0 & 3^0 \\ 3^0 & 3^0 & 3^0 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} = A$$

So the result is true for $n = 1$.

Assume the result is true for $n = k$.

$$P(k): A^k = \begin{pmatrix} 3^{k-1} & 3^{k-1} & 3^{k-1} \\ 3^{k-1} & 3^{k-1} & 3^{k-1} \\ 3^{k-1} & 3^{k-1} & 3^{k-1} \end{pmatrix}$$

Conclusion

Now we prove that the result is true for $n = k+1$.

Since,

$$\begin{aligned}
 A^{k+1} &= A \cdot A^k \\
 &= \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 3^{k-1} & 3^{k-1} & 3^{k-1} \\ 3^{k-1} & 3^{k-1} & 3^{k-1} \\ 3^{k-1} & 3^{k-1} & 3^{k-1} \end{pmatrix} \\
 &= \begin{pmatrix} 3 \cdot 3^{k-1} & 3 \cdot 3^{k-1} & 3 \cdot 3^{k-1} \\ 3 \cdot 3^{k-1} & 3 \cdot 3^{k-1} & 3 \cdot 3^{k-1} \\ 3 \cdot 3^{k-1} & 3 \cdot 3^{k-1} & 3 \cdot 3^{k-1} \end{pmatrix} \\
 &= \begin{pmatrix} 3^{(k+1)-1} & 3^{(k+1)-1} & 3^{(k+1)-1} \\ 3^{(k+1)-1} & 3^{(k+1)-1} & 3^{(k+1)-1} \\ 3^{(k+1)-1} & 3^{(k+1)-1} & 3^{(k+1)-1} \end{pmatrix}
 \end{aligned}$$

So the result is true for $n = k+1$.

Thus, by the principle of mathematical induction, we have:

$$A^n = \begin{pmatrix} 3^{n-1} & 3^{n-1} & 3^{n-1} \\ 3^{n-1} & 3^{n-1} & 3^{n-1} \\ 3^{n-1} & 3^{n-1} & 3^{n-1} \end{pmatrix}, n \in \mathbb{N}$$

Q

MISC
-3

Matrix
power by
induction



If $A = \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}$, prove that $A^n = \begin{pmatrix} 1+2n & -4n \\ n & 1-2n \end{pmatrix}$, where n is any positive integer.

Given

We are given that $A = \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}$

Substitute

We will prove the result using the principle of mathematical induction.

For $n = 1$, we have:

$$\begin{aligned}
 P(1): A^1 &= \begin{pmatrix} 1+2n & -4n \\ n & 1-2n \end{pmatrix} \\
 &= \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix} \\
 &= A
 \end{aligned}$$

So the result is true for $n = 1$.

Assume the result is true for $n = k$.

$$P(k): A^k = \begin{pmatrix} 1+2k & -4k \\ k & 1-2k \end{pmatrix}, n \in \mathbb{N}$$

Conclusion

Now we prove that the result is true for $n = k+1$.

Since,

$$\begin{aligned}
 A^{k+1} &= A \cdot A^k \\
 &= \begin{pmatrix} 1+2k & -4k \\ k & 1-2k \end{pmatrix} \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix} \\
 &= \begin{pmatrix} 3(1+2k)-4k & -4(1+2k)+4k \\ 3k+1-2k & -4k-1(1-2k) \end{pmatrix} \\
 &= \begin{pmatrix} 3+6k-4k & -4-8k+4k \\ 3k+1-2k & -4k-1+2k \end{pmatrix} \\
 &= \begin{pmatrix} 3+2k & -4-4k \\ 1+k & -1-2k \end{pmatrix} \\
 &= \begin{pmatrix} 1+2(k+1) & -4(k+1) \\ 1+k & 1-2(k+1) \end{pmatrix}
 \end{aligned}$$

So the result is true for $n = k+1$.

Thus, by the principle of mathematical induction, we have:

$$A^n = \begin{pmatrix} 1+2n & -4n \\ n & 1-2n \end{pmatrix}; n \in \mathbb{N}$$

Q

**MISC
-12**

**Commuting
matrices and
induction**



If A and B are square matrices of the same order such that $AB = BA$, then prove by induction that $AB^n = B^n A$. Further, prove that $(AB)^n = A^n B^n$ for all $n \in \mathbb{N}$.

Given

Given: A and B are square matrices of the same order such that $AB = BA$.

To prove: $P(n): AB^n = B^n A, n \in \mathbb{N}$



Substitute

For $n = 1$, we have:

$P(1): AB = BA$ [Given]

$$\Rightarrow AB^1 = B^1 A$$

So the result is true for $n = 1$.

Assume the result is true for $n = k$.

$$P(k) = AB^k = B^k A$$

Now we prove that the result is true for $n = k+1$.

$$\begin{aligned}
 AB^{k+1} &= AB^k \cdot B \\
 &= (B^k A) B \text{ [By (1)]} \\
 &= B^k (AB) \text{ [Associative law]} \\
 &= B^k (BA) \text{ [} AB = BA \text{ (Given)]} \\
 &= (B^k B) A \text{ [Associative law]} \\
 &= B^{k+1} A
 \end{aligned}$$

So the result is true for $n = k+1$.

Thus, by the principle of mathematical induction, we have $AB^n = B^n A, n \in \mathbb{N}$

Now we prove that $(AB)^n = A^n B^n$ for all $n \in \mathbb{N}$

For $n = 1$, we have:

$$(AB)^1 = A^1 B^1 = AB$$

So the result is true for $n = 1$.

Assume the result is true for $n = k$.

$$(AB)^k = A^k B^k$$

Conclusion

Now we prove that the result is true for $n = k+1$.

$$\begin{aligned} A B^{k+1} &= (A B)^k \cdot (A B) \\ &= (A^k B^k) \cdot (A B) \text{ [By (2)]} \\ &= A^k (B^k A) B \text{ [Associative law]} \\ &= A^k (A B^k) B \text{ [} A B^n = B^n A, n \in \mathbb{N} \text{]} \\ &= (A^k A) \cdot (B^k B) \text{ [Associative law]} \\ &= A^{k+1} B^{k+1} \end{aligned}$$

So the result is true for $n = k+1$.

Thus, by the principle of mathematical induction, we have $(A B)^n = A^n B^n, n \in \mathbb{N}$